



Samsung Telecommunications America Product & User Experience Research

2010 Research Reports Overview

Date: 02-24-2012

Design & Technology PUX Research

Strategic Marketing Research

Samsung Telecommunications America

Vision

**Create and sustain an open, collaborative, transparent,
and unbiased user/product research culture for informed
business decisions and future design strategy**

Mission Statement

Provide informative, scientific, and actionable results for innovation and/or improvement of our product/service design & features through world-class product /user research practices.

Goals

Understand users' needs and contexts of use to inform future innovative product/service concepts & UX/Ergonomics design strategy.

Validate & improve UX/Ergonomics design through iterative user/product research for key strategic products/services

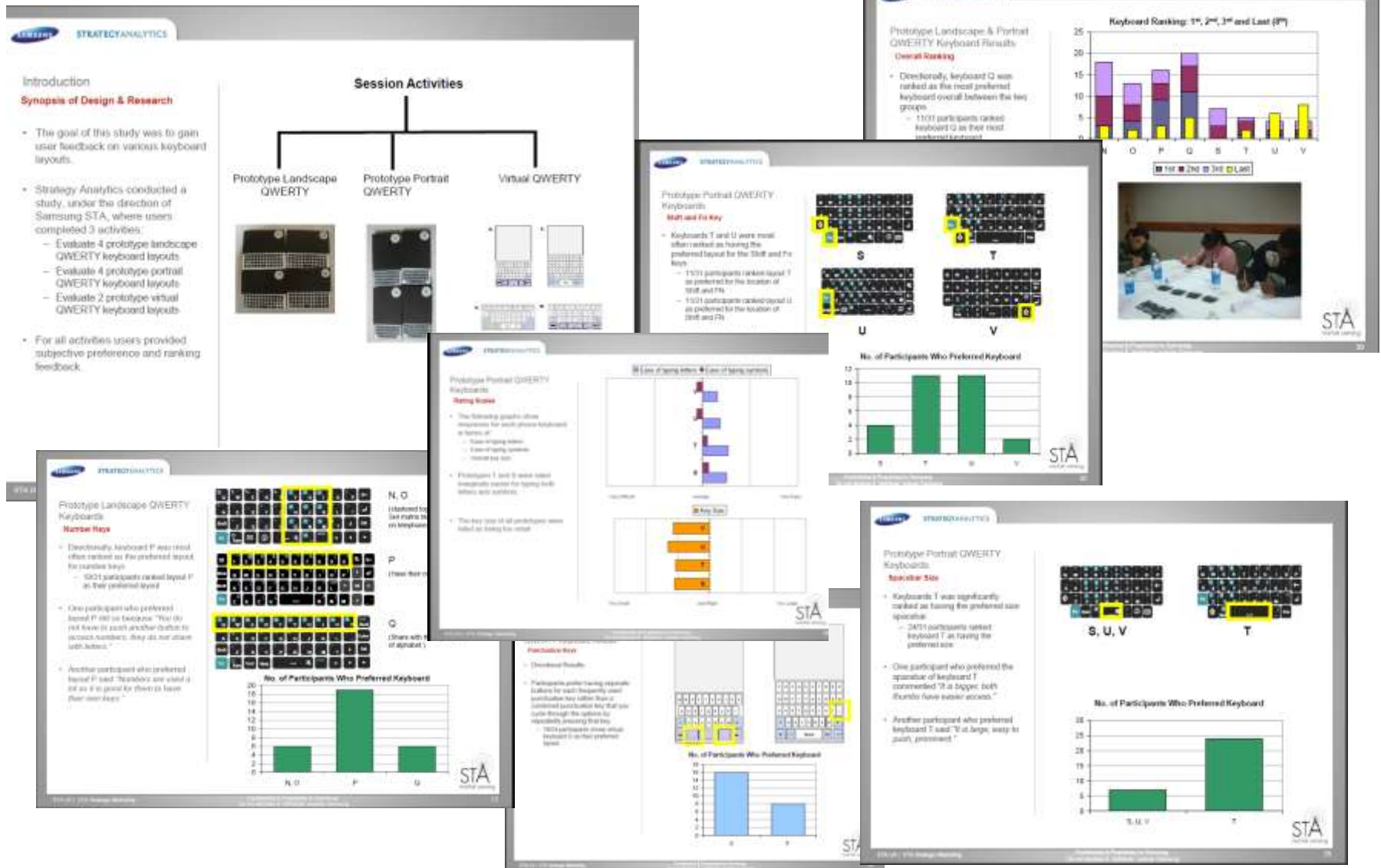
Anticipate latent and future user, market, & product evolution through proactive user and market sensing.

Chart competitive UX/Design Landscape; Situate and validate sustainable innovation and differentiation in the market.

Research ideation and initiation, needs/questions come from our team and different organizations -HQ Design Group, Design & Technology, Marketing, Product Planning, Product Management, HQ Product Planning, Carrier UX teams.

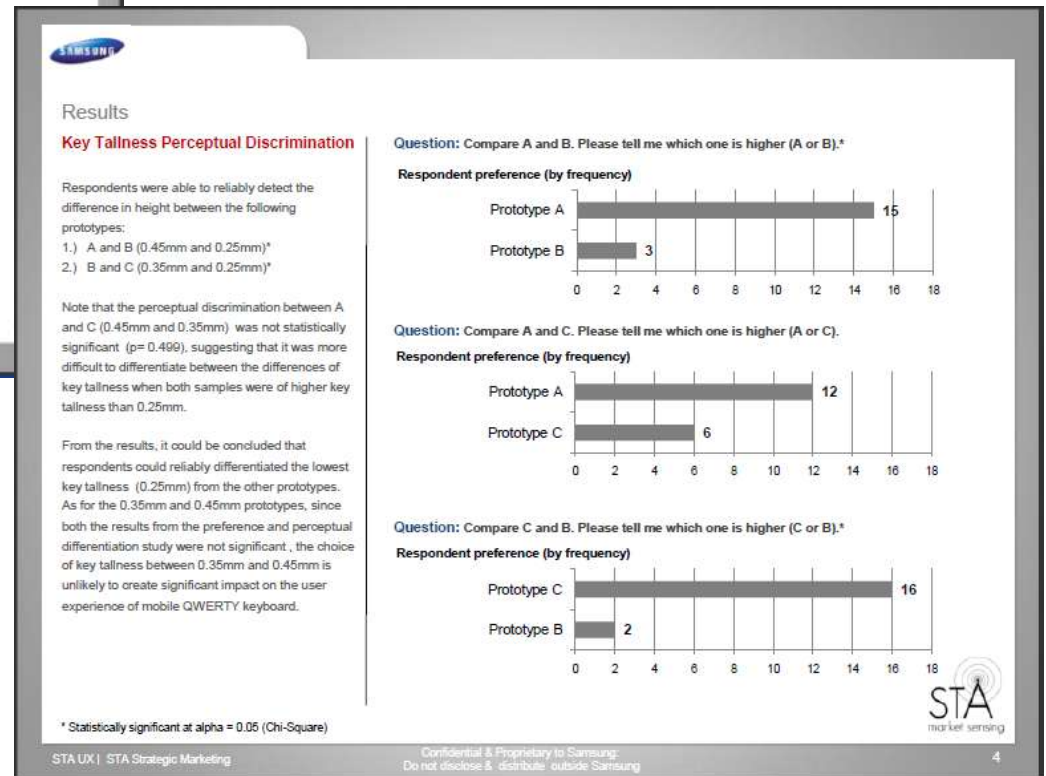
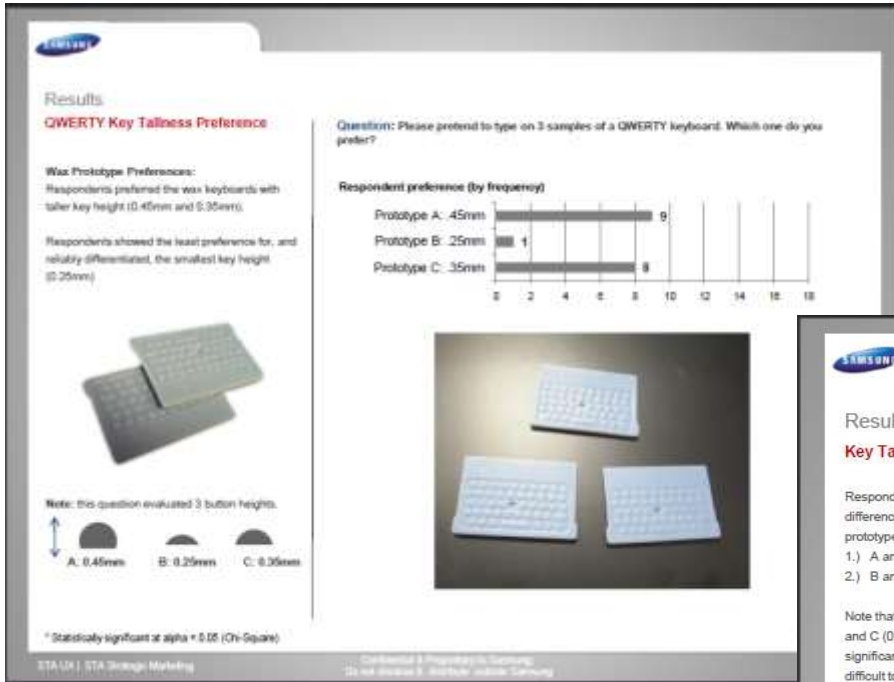
1. QWERTY Prototype Keyboard Layout Evaluation

Overview and Representative Results



2. Key Tallness User Discrimination and Preference Study

Overview and Results



3. Comparative Visual Design Style: Samsung/Android Theme Color User Preference Study

Overview and Representative Results

Introduction
Synopsis of Design & Research

- The goal of this study was to gather user preference of 5 different styles of visual design for Samsung Android touchscreen, Settings screen, Contacts screen, Notification Window and general dialing box. This study took place in San Diego as part of a larger study for different topics between Feb 8th and 11th run jointly by STA UK and Strategy Analytics.
- Respondents were asked to rate their preference of 5 different styles.
- For the most and least favorite visual design, respondents used 2 adjectives to describe the feeling evoked by the design.

Session Activities

1 Moderator
 1-1 Respondents

Preference Ranking and Adjectives Description

5 different styles/themes

STA UK | STA Strategy Analytics

Results
Gender Differences

- Overall, female respondents preferred B-2, B-3, A-1, A-2 and B-1. The results suggested that B-2, B-3 and A-1 as a group can be selected as prospective candidates for the themes available.
- Male respondents preferred A-1 and B-2, which were also preferred by female respondents as a group.

Female only Theme

Sample	Frequency	Sum of ranks	Mean of ranks	Groups
B-2	18	34,000	1.889	A
B-3	18	42,000	2.333	A
A-1	18	47,000	2.611	A
A-2	18	71,000	3.944	B
B-1	18	76,000	4.222	B

Female Ranking Preference

Sample	Frequency	Sum of ranks	Mean of ranks	Groups
A-1	20	27,000	1.350	A
B-2	20	48,000	2.300	B
A-2	20	70,000	3.500	C
B-1	20	75,000	3.750	C

STA UK | STA Strategy Analytics

Results
Overall Ranking

- Results (N=88) were statistically significant with Style A-1 and B-2 ranked higher than the rest across all respondents.
- Reactions to some of the designs were polarized along gender (B-3 was ranked second in female compared to last in male).
- Typical adjectives used to describe the most favorite designs were "bright", "colorful", "clean", "simple".
- Typical adjectives used to describe the least favorite designs were "dark", "dull", "boring".

Sample	Frequency	Sum of ranks	Mean of ranks	Groups
A-1	38	74,000	1.947	A
B-2	38	80,000	2.105	A
B-3	38	124,000	3.263	B
A-2	38	141,000	3.711	B, C
B-1	38	151,000	3.974	C

Top ranked Themes

Bottom ranked Theme

STA UK | STA Strategy Analytics

Results
Ranking of B2

Gender-Safe design

Style B-2

Number of respondents

Rank (1st highest, 5th lowest)

Female (red bar), Male (blue bar)

STA UK | STA Strategy Analytics

4. Comparative UX: Optical Joystick/ Physical Direction Pad Navikey(Performance & Usability)

Overview and Representative Results

Executive Summary
Optical Mouse versus D-Pad Navigation

The purpose of this study is to investigate differences in user experience (performance and usability) of optical joystick and physical direction pad using a variety of content types to include document.

The study includes, scrolling and highlighting, larger navigation to access to home screen, set of home, menu grid, and a spreadsheet (drag and drop) as well as web browsing and gaming using both navigation controls.

Hypotheses:
There is no significant difference in usability and feature performance across a set of typical tasks (document scrolling, menu scrolling, menu grid navigation, web browsing, game) using either optical joystick or physical direction pad ("4-way" button).

Results:
All respondents (22/27) preferred D-Pad to Optical Mouse. Respondents rated D-Pad significantly better, and did not rate accurate the optical mouse. Performance of using D-Pad was significantly faster than Optical joystick, and was significantly better in accuracy, speed and help scrolling.



To ensure valid comparisons, Nokia E72 was used as it supports both optical joystick and D-Pad as Navikey.

Summary of Findings: D-Pad and Optical Joystick Performance & Rating

Overall, all respondents preferred D-Pad navigation over the Optical Mouse. This respondent reported that they would like to purchase a laptop with optical joystick because it would make his work faster, more accurate and business-like, even though the preferred D-Pad as navigation device.

Another respondent rated some of the tasks faster using Optical joystick but preferred D-Pad. The reason he gave was that he had a large number of songs in his device and he knew it would take substantially longer to scroll a large list.

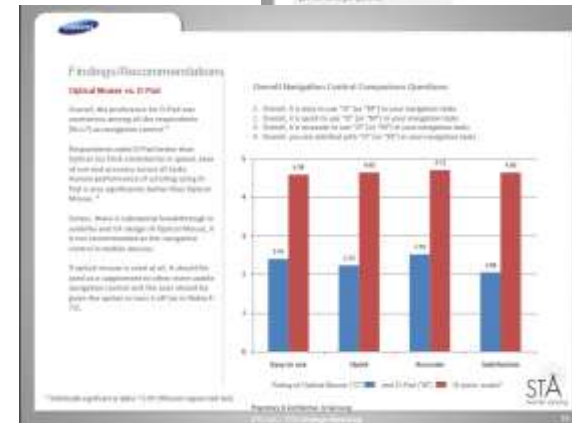
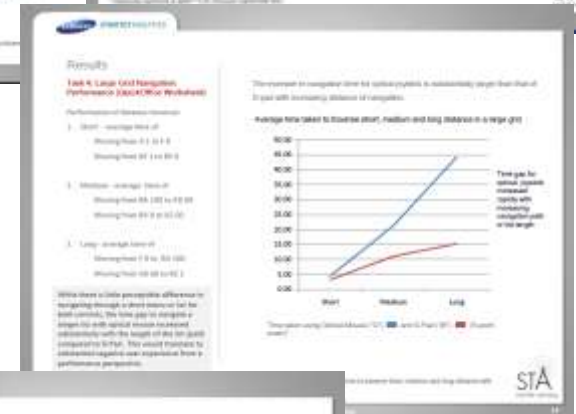
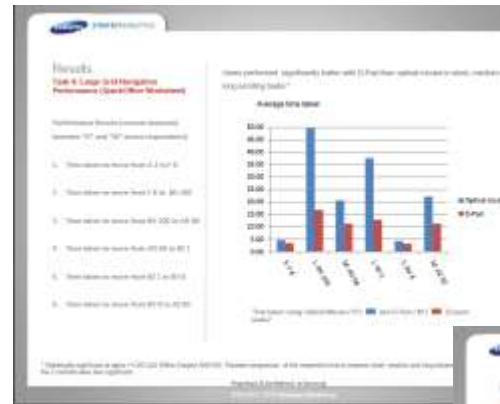
Respondents also generally said D-Pad is better than Optical Mouse for navigation. The most important reason was that he had a large number of songs in his device and he knew it would take substantially longer to scroll a large list.

Respondents also generally said D-Pad is better than Optical Mouse for navigation. The most important reason was that he had a large number of songs in his device and he knew it would take substantially longer to scroll a large list.

Typical Comments:
Optical Joystick: "Not for finger" (Usability), "I played music and used to physical key" (Usability), "Not much" (Usability), "The feedback is important to me" (Usability).
Optical Joystick: "Not for finger" (Usability), "I played music and used to physical key" (Usability), "Not much" (Usability), "The feedback is important to me" (Usability).

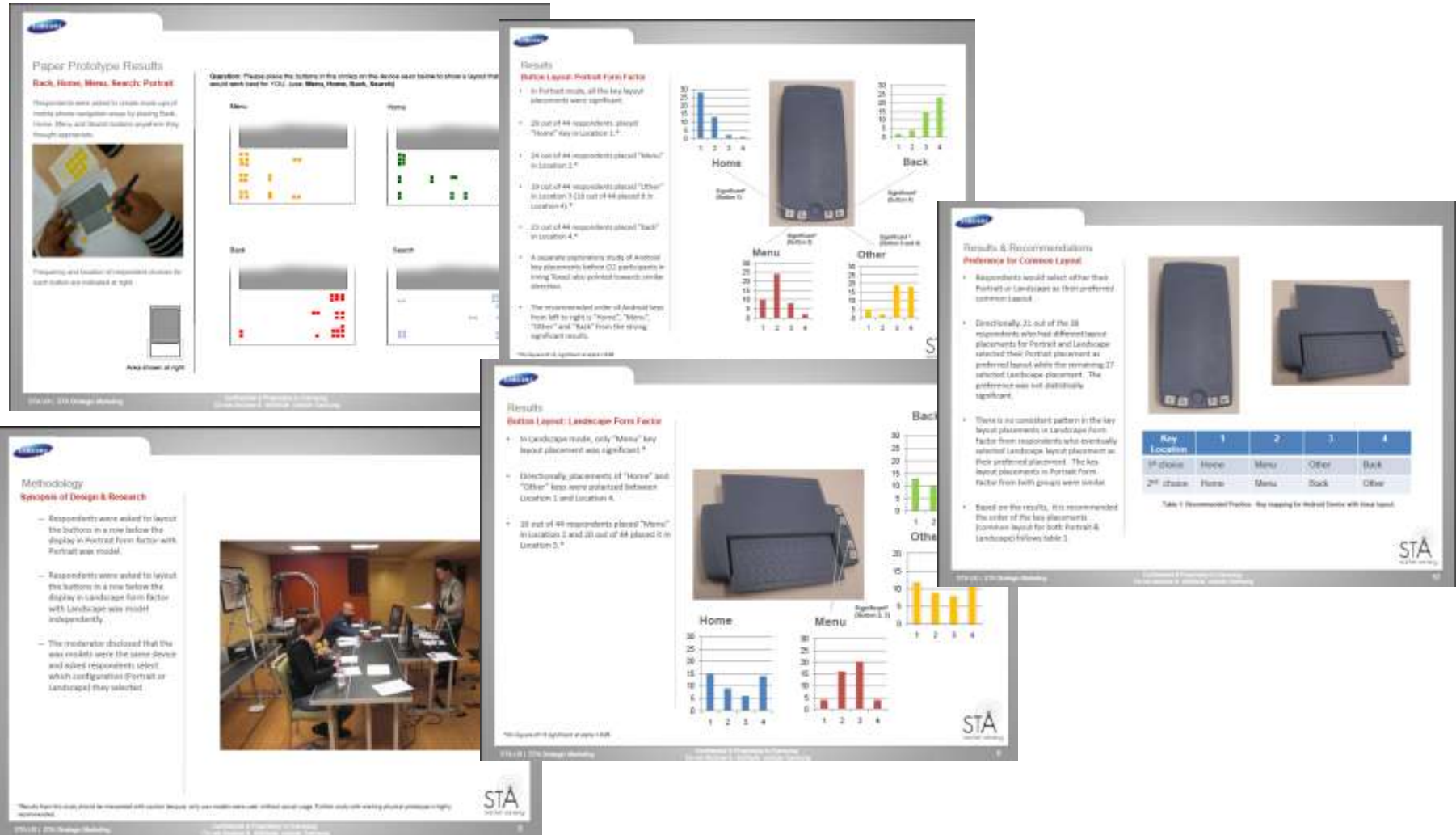
Recommendations:
Recommendation: D-Pad is better than Optical Mouse for navigation. The most important reason was that he had a large number of songs in his device and he knew it would take substantially longer to scroll a large list.

Performance & Usability:
Performance & Usability: D-Pad is better than Optical Mouse for navigation. The most important reason was that he had a large number of songs in his device and he knew it would take substantially longer to scroll a large list.



5. Android Hard Keys Layout: Portrait, Landscape and Common Key Mappings

Overview and Representative Results

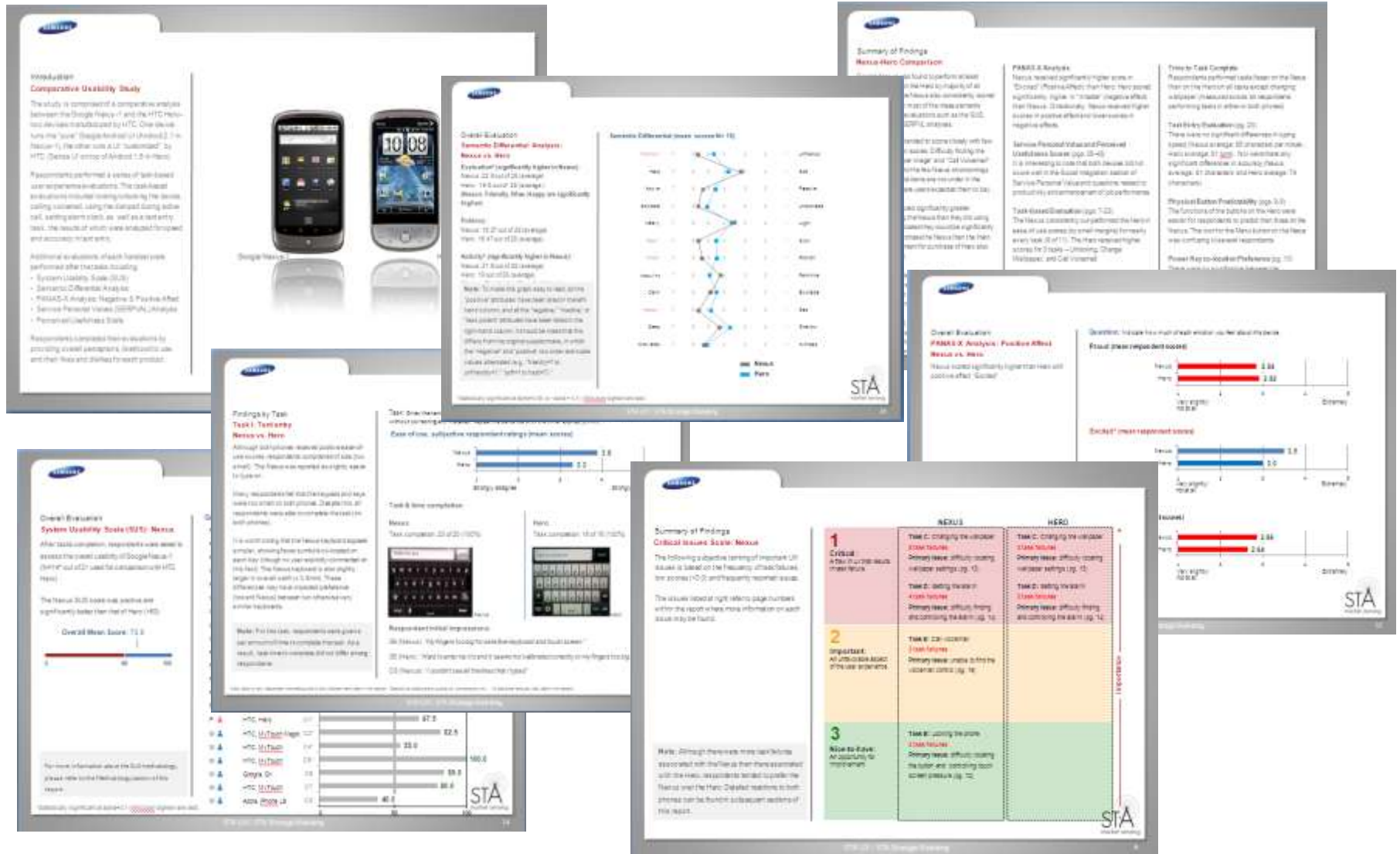


6. Competitive Android UX: Dialer & Contacts Samsung S1 Prototype, Google Nexus-1 & HTC Hero

Overview and Representative Results

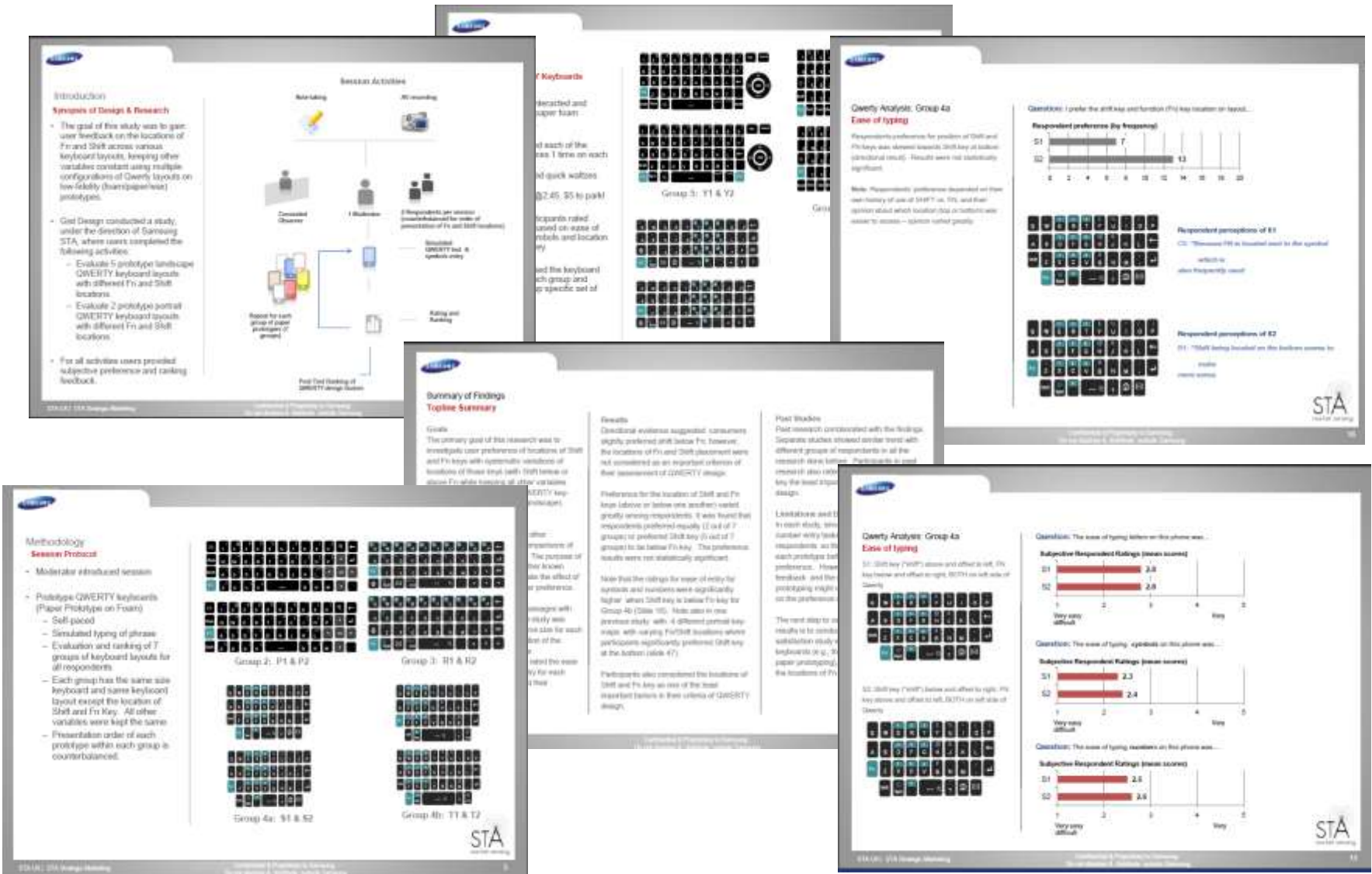


7. Comparative Android User Experience: Google Nexus-1 vs. HTC Hero



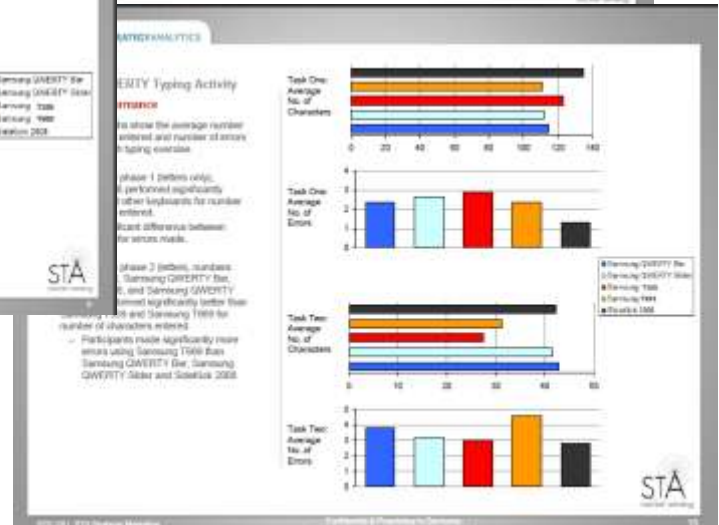
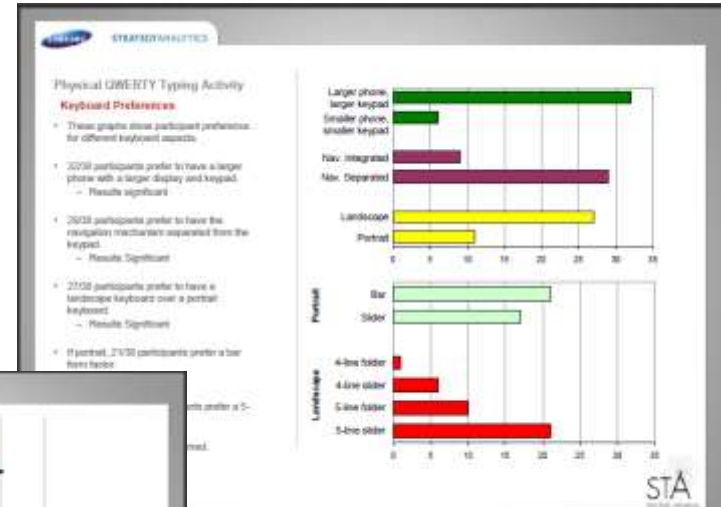
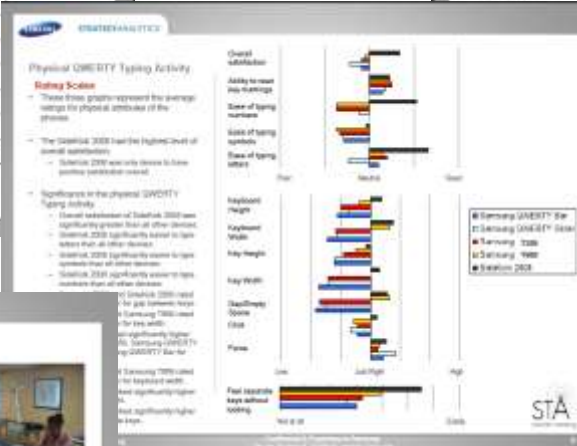
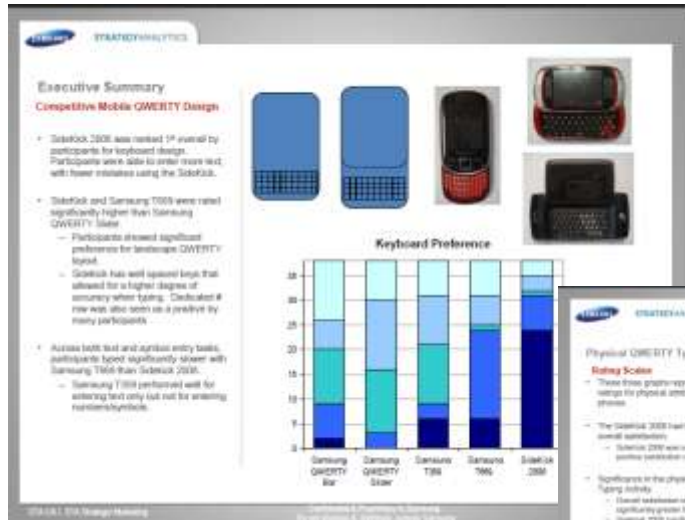
8. Mobile QWERTY Keyboard: Shift/Fn Locations (controlled study using paper prototypes)

Overview and Representative Results

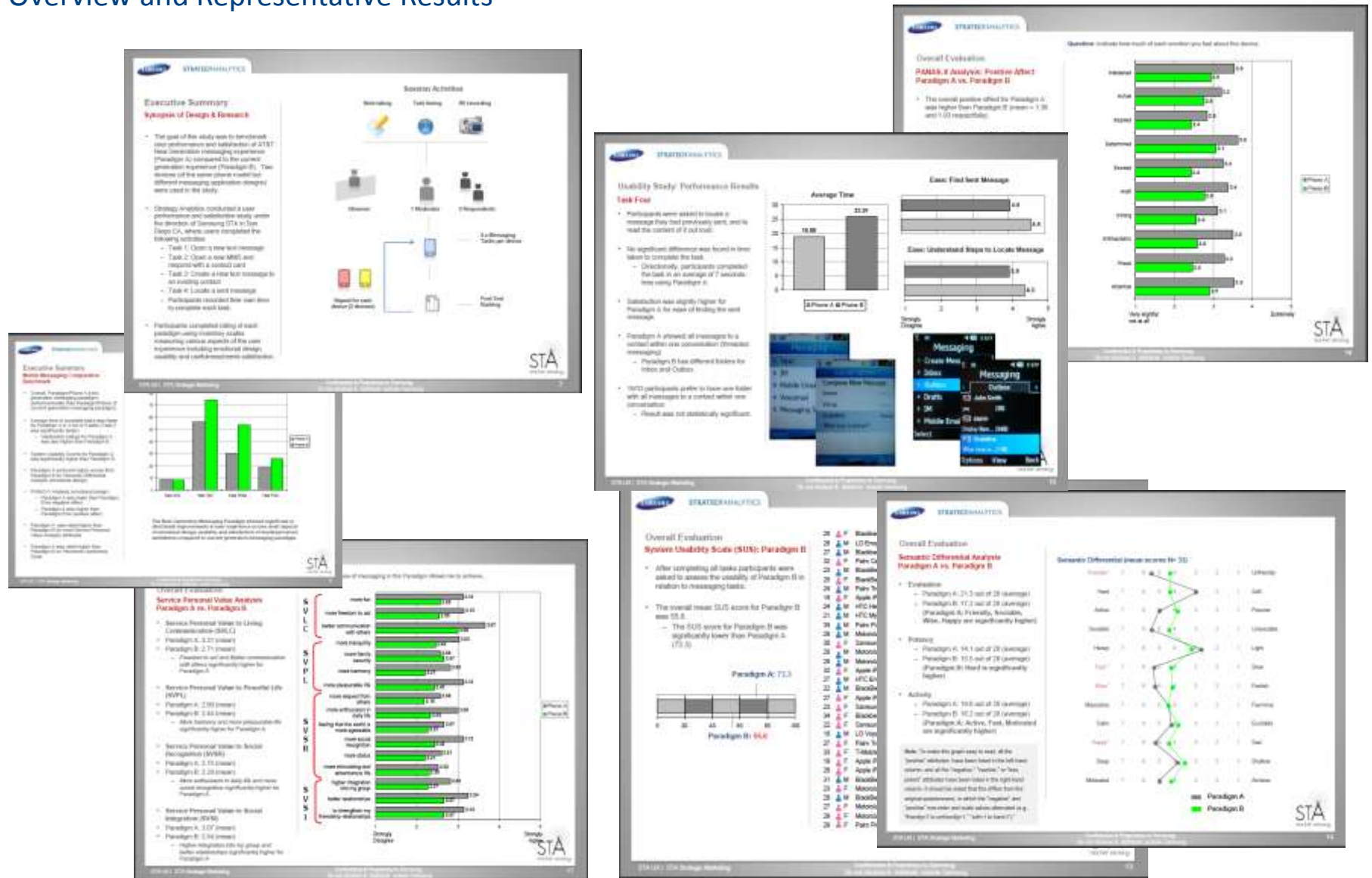


9. Competitive QWERTY Performance & Satisfaction Benchmark (Sidekick 2008, and 4 Samsung Pre-Launch Devices)

Overview and Representative Results

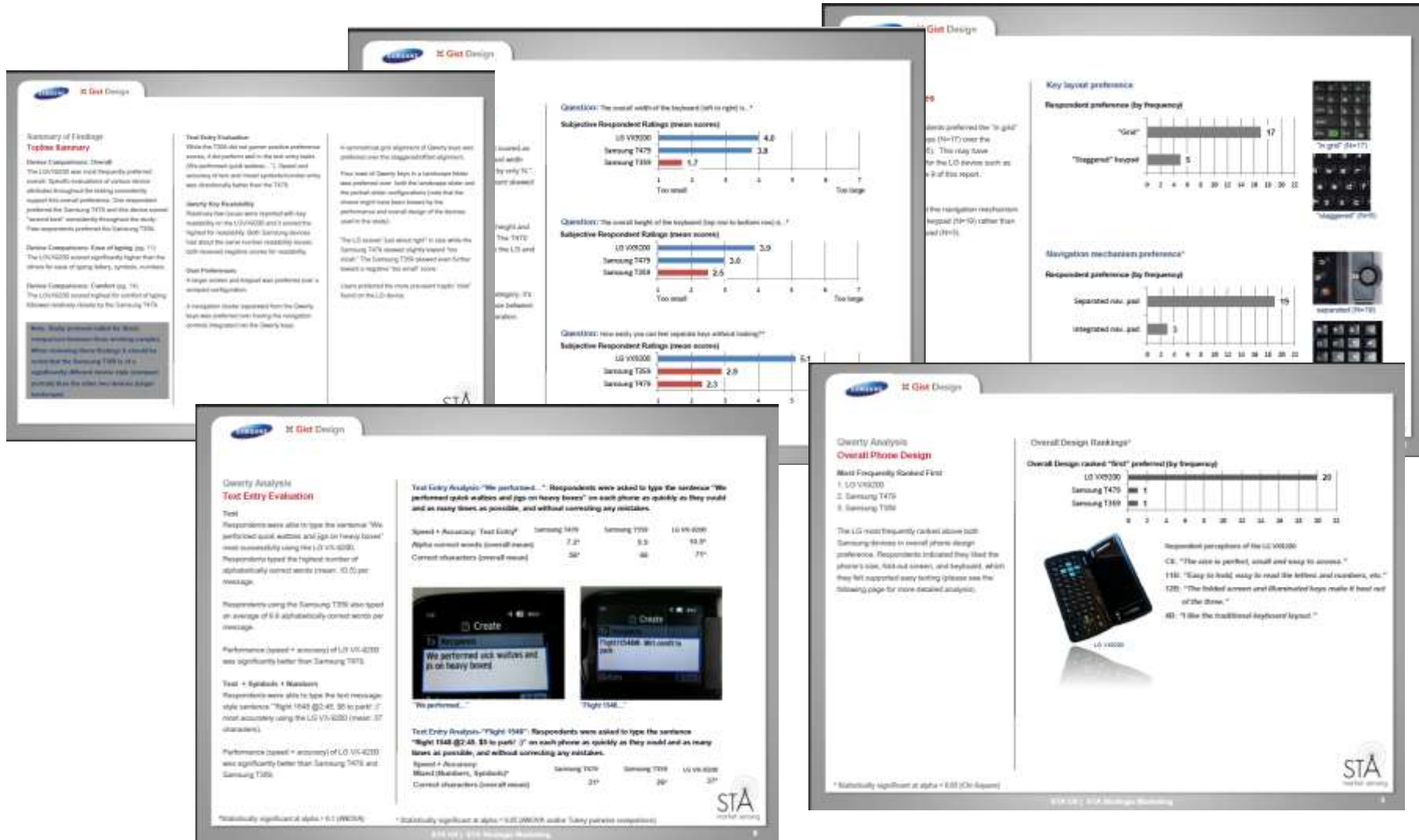


10. Mobile Messaging : Comparative AT&T Next & Current Generation Messaging Paradigms



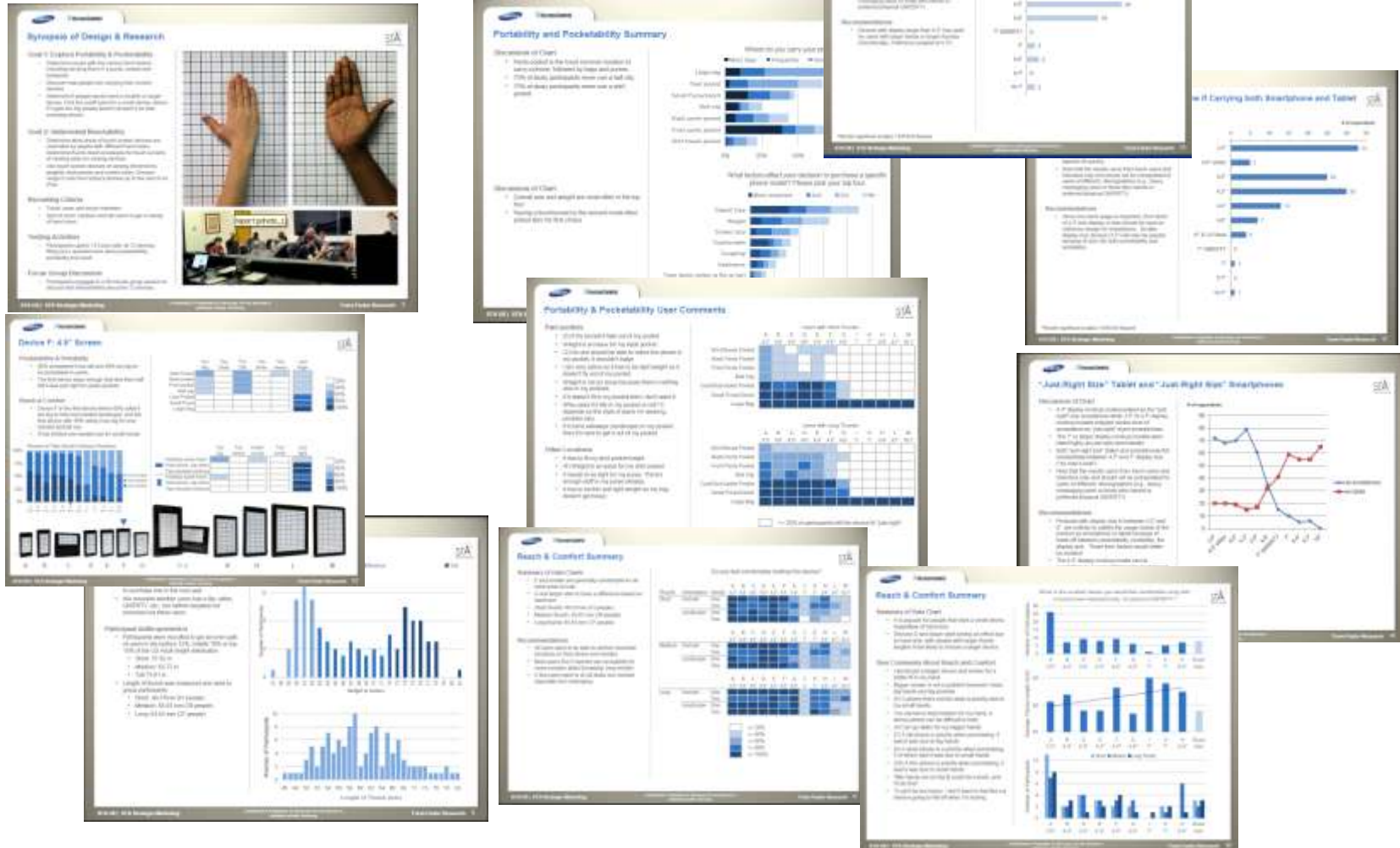
11. Competitive QWERTY Performance & Satisfaction Benchmark (LG Env3, and 2 Samsung Pre-Launch Devices, T359 and T479)

Overview and Representative Results



12. Form factor Research: devices from 3.5" display to 10" display

Overview and Representative Results



Overview and Representative Results

[illegible][illegible]

Software Concept Evaluation

Smart Value Engineering

Examples

- Comparison of three cases in order of preference: Usability

- Relative frequency for each quality
- Priority for usability for all scenarios
- Priority for usability in the scenario
- Relative usability in the scenario
- Relative usability

Relative Usability

Example 2

- Basic design comparison between two cases, comparing the quality of the design with the quality of the design
- Comparison of the quality of the design with the quality of the design
- Comparison of the quality of the design with the quality of the design

Relative Quality of Design

- Comparison of the quality of the design with the quality of the design
- Comparison of the quality of the design with the quality of the design
- Comparison of the quality of the design with the quality of the design

Example 3

- Comparison of the quality of the design with the quality of the design
- Comparison of the quality of the design with the quality of the design
- Comparison of the quality of the design with the quality of the design

Relative Quality

Innovative
Useful

Useful

Cost

Fun

Security

Quality

Design

Design

University of Applied Sciences

University of Applied Sciences

University of Applied Sciences

Physical Form Factor Evaluation

Language Fig. 22

Overview

- 1. Ask customer what he wants, describe existing
- 2. Ask customer what he needs
 - Describe existing mobility for each device
 - Use it to specify what he needs
- 3. Consider it a goal for the partnership. Don't forget!
- 4. The longer it takes to build, the more willing are the clients will be (even though) the smaller laptop
- 5. Customer must perceive that they're a friend
- 6. Represent agency capabilities for productivity
- 7. The longer it takes to build, the more he values a product made

The chart shows two bars for 'Customer Value'. The first bar, 'Time with a new product', is significantly higher than the second bar, 'Time with a new product'. The y-axis is labeled 'Customer Value' and ranges from 0 to 100.

Recommendations

- 1. Represent a mobile device
- 2. Provide a design for the customer looking for a device to use the device for a long time
- 3. Represent a device for the long term device
- 4. Represent a device for the long term device
- 5. Represent a device for the long term device
- 6. Represent a device for the long term device
- 7. Represent a device for the long term device
- 8. Represent a device for the long term device
- 9. Represent a device for the long term device
- 10. Represent a device for the long term device
- 11. Represent a device for the long term device
- 12. Represent a device for the long term device
- 13. Represent a device for the long term device
- 14. Represent a device for the long term device
- 15. Represent a device for the long term device
- 16. Represent a device for the long term device
- 17. Represent a device for the long term device
- 18. Represent a device for the long term device
- 19. Represent a device for the long term device
- 20. Represent a device for the long term device
- 21. Represent a device for the long term device
- 22. Represent a device for the long term device
- 23. Represent a device for the long term device
- 24. Represent a device for the long term device
- 25. Represent a device for the long term device
- 26. Represent a device for the long term device
- 27. Represent a device for the long term device
- 28. Represent a device for the long term device
- 29. Represent a device for the long term device
- 30. Represent a device for the long term device
- 31. Represent a device for the long term device
- 32. Represent a device for the long term device
- 33. Represent a device for the long term device
- 34. Represent a device for the long term device
- 35. Represent a device for the long term device
- 36. Represent a device for the long term device
- 37. Represent a device for the long term device
- 38. Represent a device for the long term device
- 39. Represent a device for the long term device
- 40. Represent a device for the long term device
- 41. Represent a device for the long term device
- 42. Represent a device for the long term device
- 43. Represent a device for the long term device
- 44. Represent a device for the long term device
- 45. Represent a device for the long term device
- 46. Represent a device for the long term device
- 47. Represent a device for the long term device
- 48. Represent a device for the long term device
- 49. Represent a device for the long term device
- 50. Represent a device for the long term device
- 51. Represent a device for the long term device
- 52. Represent a device for the long term device
- 53. Represent a device for the long term device
- 54. Represent a device for the long term device
- 55. Represent a device for the long term device
- 56. Represent a device for the long term device
- 57. Represent a device for the long term device
- 58. Represent a device for the long term device
- 59. Represent a device for the long term device
- 60. Represent a device for the long term device
- 61. Represent a device for the long term device
- 62. Represent a device for the long term device
- 63. Represent a device for the long term device
- 64. Represent a device for the long term device
- 65. Represent a device for the long term device
- 66. Represent a device for the long term device
- 67. Represent a device for the long term device
- 68. Represent a device for the long term device
- 69. Represent a device for the long term device
- 70. Represent a device for the long term device
- 71. Represent a device for the long term device
- 72. Represent a device for the long term device
- 73. Represent a device for the long term device
- 74. Represent a device for the long term device
- 75. Represent a device for the long term device
- 76. Represent a device for the long term device
- 77. Represent a device for the long term device
- 78. Represent a device for the long term device
- 79. Represent a device for the long term device
- 80. Represent a device for the long term device
- 81. Represent a device for the long term device
- 82. Represent a device for the long term device
- 83. Represent a device for the long term device
- 84. Represent a device for the long term device
- 85. Represent a device for the long term device
- 86. Represent a device for the long term device
- 87. Represent a device for the long term device
- 88. Represent a device for the long term device
- 89. Represent a device for the long term device
- 90. Represent a device for the long term device
- 91. Represent a device for the long term device
- 92. Represent a device for the long term device
- 93. Represent a device for the long term device
- 94. Represent a device for the long term device
- 95. Represent a device for the long term device
- 96. Represent a device for the long term device
- 97. Represent a device for the long term device
- 98. Represent a device for the long term device
- 99. Represent a device for the long term device
- 100. Represent a device for the long term device

The chart shows two bars for 'Customer Value'. The first bar, 'Time with a new product', is significantly higher than the second bar, 'Time with a new product'. The y-axis is labeled 'Customer Value' and ranges from 0 to 100.

Physical Product Evaluation

Multi-tasking	Personal: review and reformatting/responses; email and review a merge Productivity: phone call and contact/communication; email and Word's PDF attachment Students: Productivity: document preparation and format/proofreading/extra
Streamline views	Personal: travel in a city (street view is less useful in suburban/rural areas) Productivity: discover to a meeting/business trip destination Alternative views suggested by users: 1) on-line map and turn-by-turn, on-line map and back-to-back map
Enhanced views	Productivity: email and reply, list and messages, two parts of a document, subeditor/contact and detail reference/reference view and details (e.g. medical/diagnosis) Games: multi-player games, chat view games Students: textbook page and pieces of a syllabus
Real-time conferencing	Personal: family reunion (relatives, members of the armed services) where live events (e.g. family reunion) can be viewed Business: video conference meetings MMS: they can learn with friends
Advanced messaging	MMS: view conversation and related photos/videos Alternative applications suggested by users: Facebook is especially important, Twitter, email
Real-time updates	Students: textbook page and distribution, non-adjusted pages, search and search results Personal: table of contents and internet page, page and distribution, textbook and internet page
Open third browser	MMS: Facebook friend and posts on a photo/video Personal: comparison shopping, photo/collaboration
Desktop	Personal: comparison shopping, trip planning, medical information

Screen & Mobile Navigation

Quick and convenient

- Flexible use scenario of mobile navigation
- Use as a guide or as a navigation tool
- The system can be used in a variety of ways
- The system can be used in a variety of ways
- The system can be used in a variety of ways

Navigation

- The system can be used in a variety of ways
- The system can be used in a variety of ways

Screen

- The system can be used in a variety of ways
- The system can be used in a variety of ways

Recommendations

- The system can be used in a variety of ways
- The system can be used in a variety of ways



Innovative
Useful
Simple
Easy
Light
Portable
Quick
Versatile



Navigation App



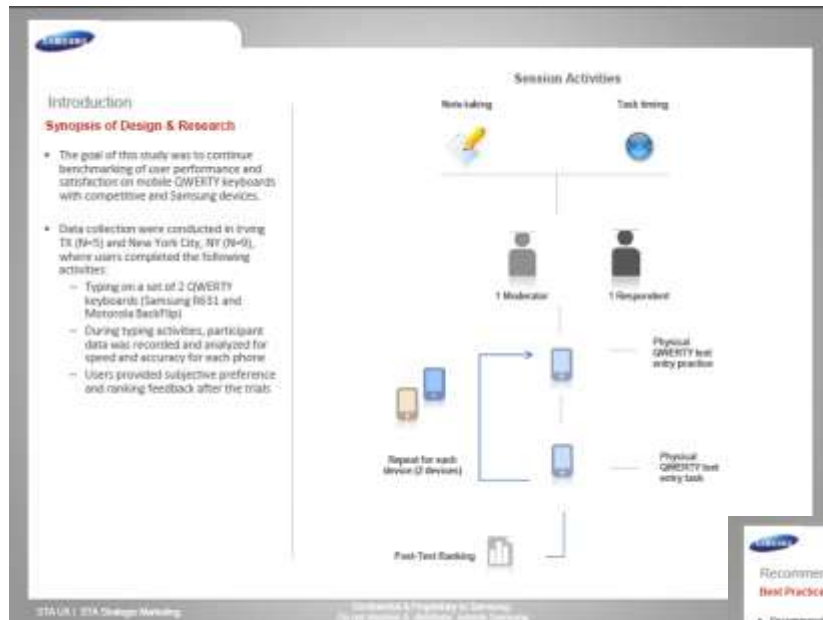
Navigation App



Navigation App

15. QWERTY Design: Comparative User Performance & Satisfaction Benchmark (Samsung R631 and Motorola BackFlip)

Overview and Representative Results



Executive Summary
Competitive Mobile QWERTY Design

- A competitive usability (user performance & satisfaction) study of mobile QWERTY keyboards was conducted with Samsung R631 and Motorola BackFlip (N=14).
- Samsung R631 was rated significantly lower in key gap, key width and height, keyboard width and height and overall size to type comfortably.
- There were no significant differences in typing performance (both speed and error) of text entry and mixed text/number/symbols entry between Motorola BackFlip and Samsung R631.
- There were no significant differences in the subjective satisfaction ratings between Motorola Flip and Samsung R631.
- There were no significant differences in the ranking / preference of QWERTY design and overall design between the devices.

- Qualitative feedback from respondents also corroborated that the key sizes and the keyboard of R631 were perceived to be small for comfortable typing.
- The key height of BackFlip (8mm) is approximately 1.7 times that of Samsung R631 (4.6mm), making the keys appear to be much larger both in width and height.
- It is interesting to note that the physical measurements of the key width and key gap of Samsung R631 and Motorola BackFlip are approximately the same (7.5 mm, 0.7mm) while respondents perceived the key width and key gap of R631 to be significantly smaller.
- The overall size of BackFlip is only slightly longer (about 3mm) and the differences of width and depth between the devices are around 10mm.

- Recommended key height for landscape device is greater than 5.5mm (minimum). See pg. 12 from Mobile Keyboard Best Practice Final Research Report & Physical Design Variables Recommendations (Oct 09)

BackFlip
Approximate Size of device: 105mm x 54mm x 15mm
Trip force: around 200gf

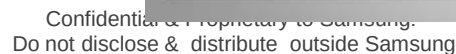
R631
Approximate Size of device: 105mm x 54mm x 15mm
Trip force: around 200gf

Recommendations
Best Practice Mobile QWERTY Design

- Recommended key height for landscape device is greater than 5.5mm.
- For the design of future mobile QWERTY keyboards, we would recommend following the best practices established from user research. The current recommendations can be found in "Mobile QWERTY Keyboard Best Practice Final Research Report & Physical Design Variables Recommendations" dated 10-14-09.
- The recommendations will be updated later this year pending additional user research on both physical and virtual keyboard design.

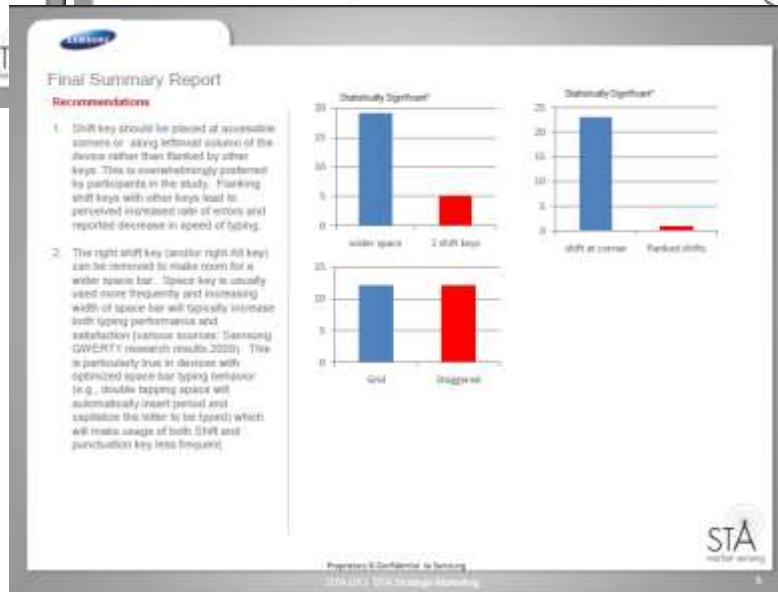
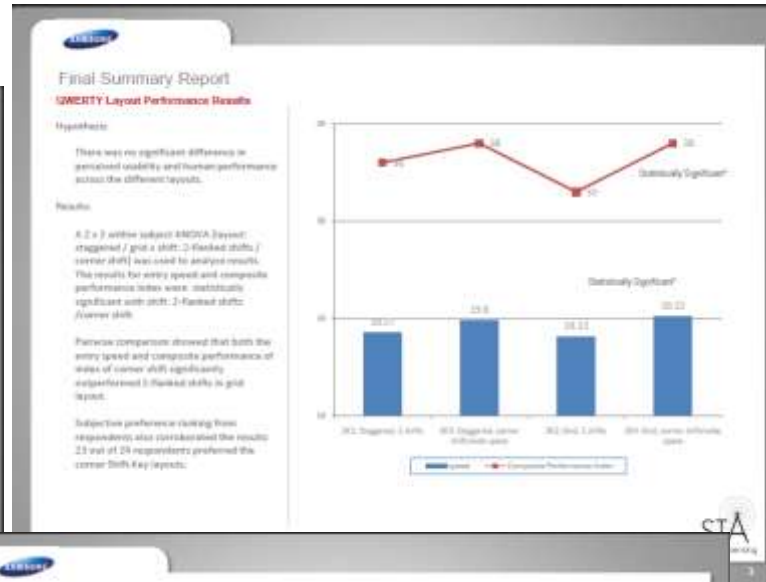
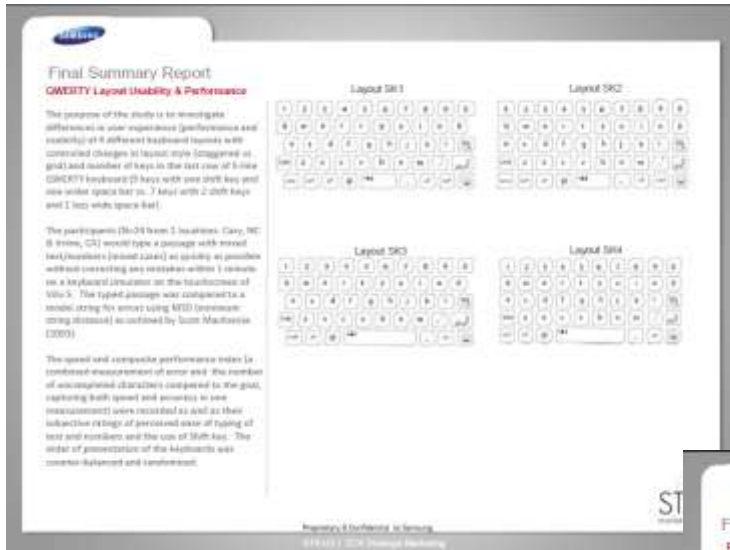
STA UI/UX | STA Strategic Marketing
Confidential & Proprietary to Samsung
Do not disclose & distribute outside Samsung

SAMSUNG



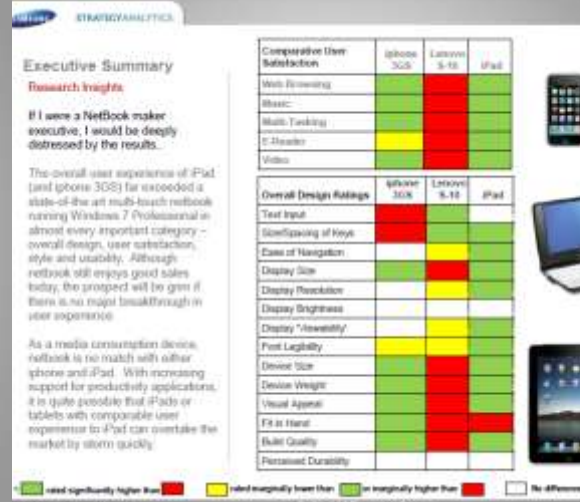
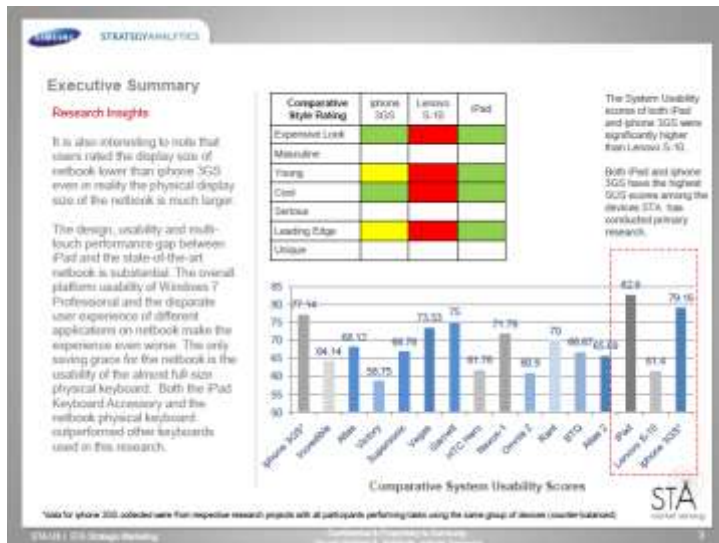
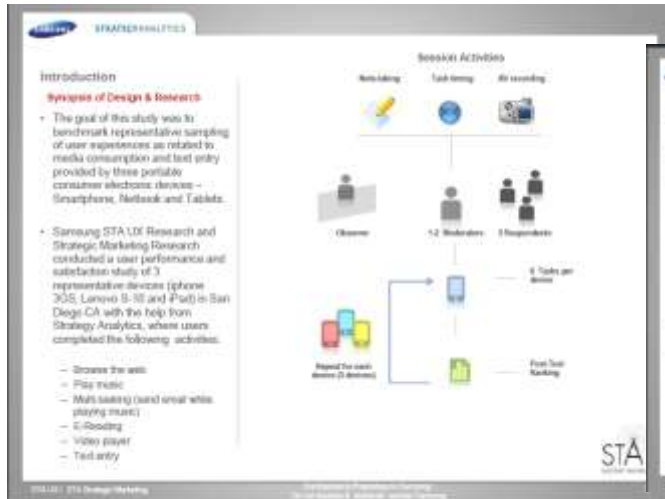
17. Comparative QWERTY Layout Performance and Usability(Variations from Sidekick baseline)

Overview and Representative Results



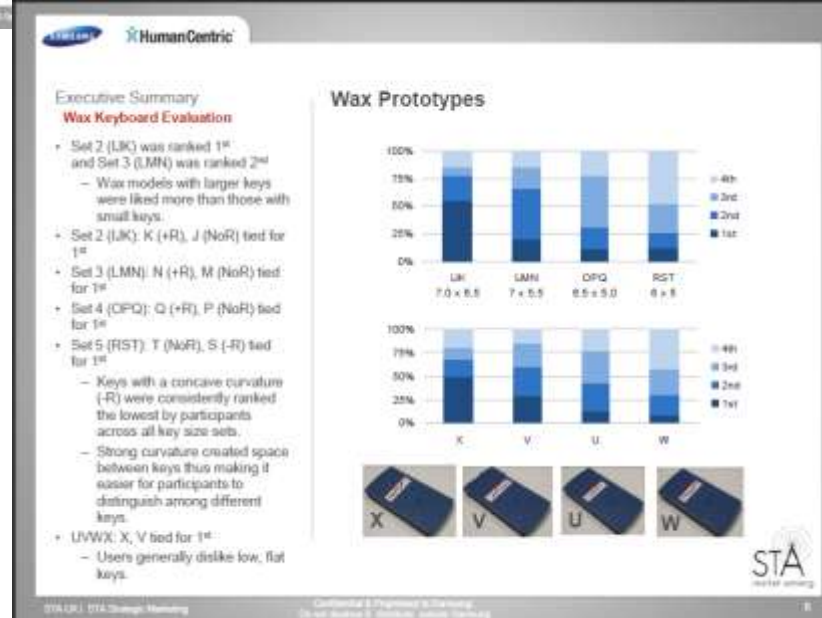
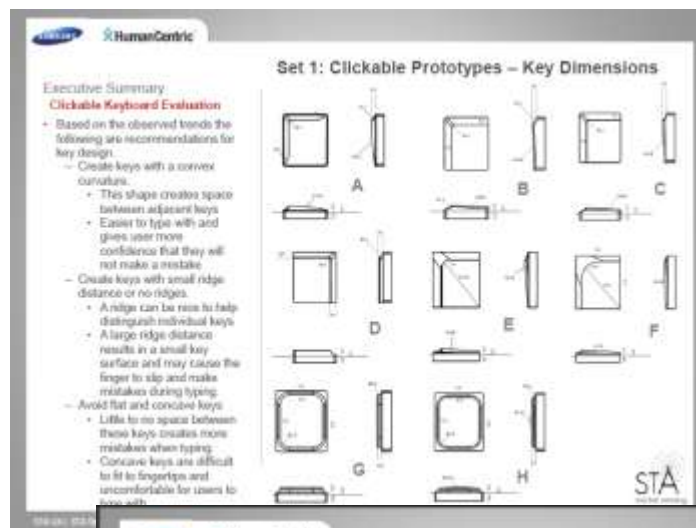
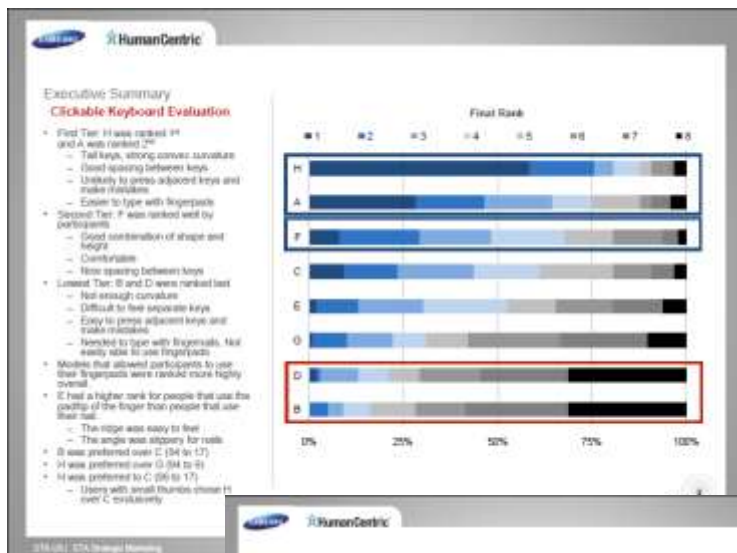
18. Competitive Smartphone, Netbook and Tablet: UX, Design, Media Consumption, Accessories (iPhone 3GS, Lenovo S-10 & iPad)

Overview and Representative Results

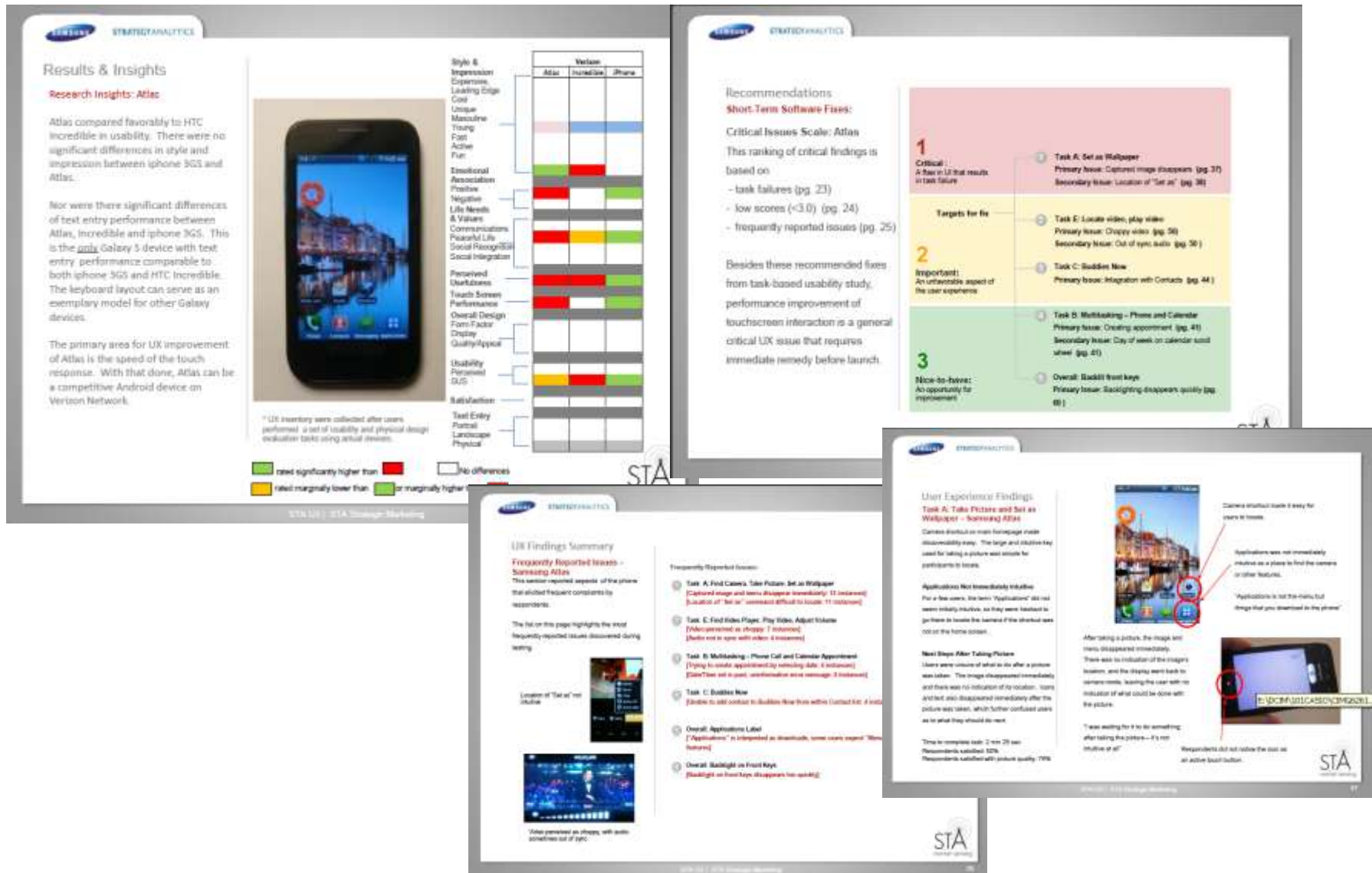


19. Mobile QWERTY Keypad Key Shape/Curvature (with HQ Design Group)

Overview and Representative Results

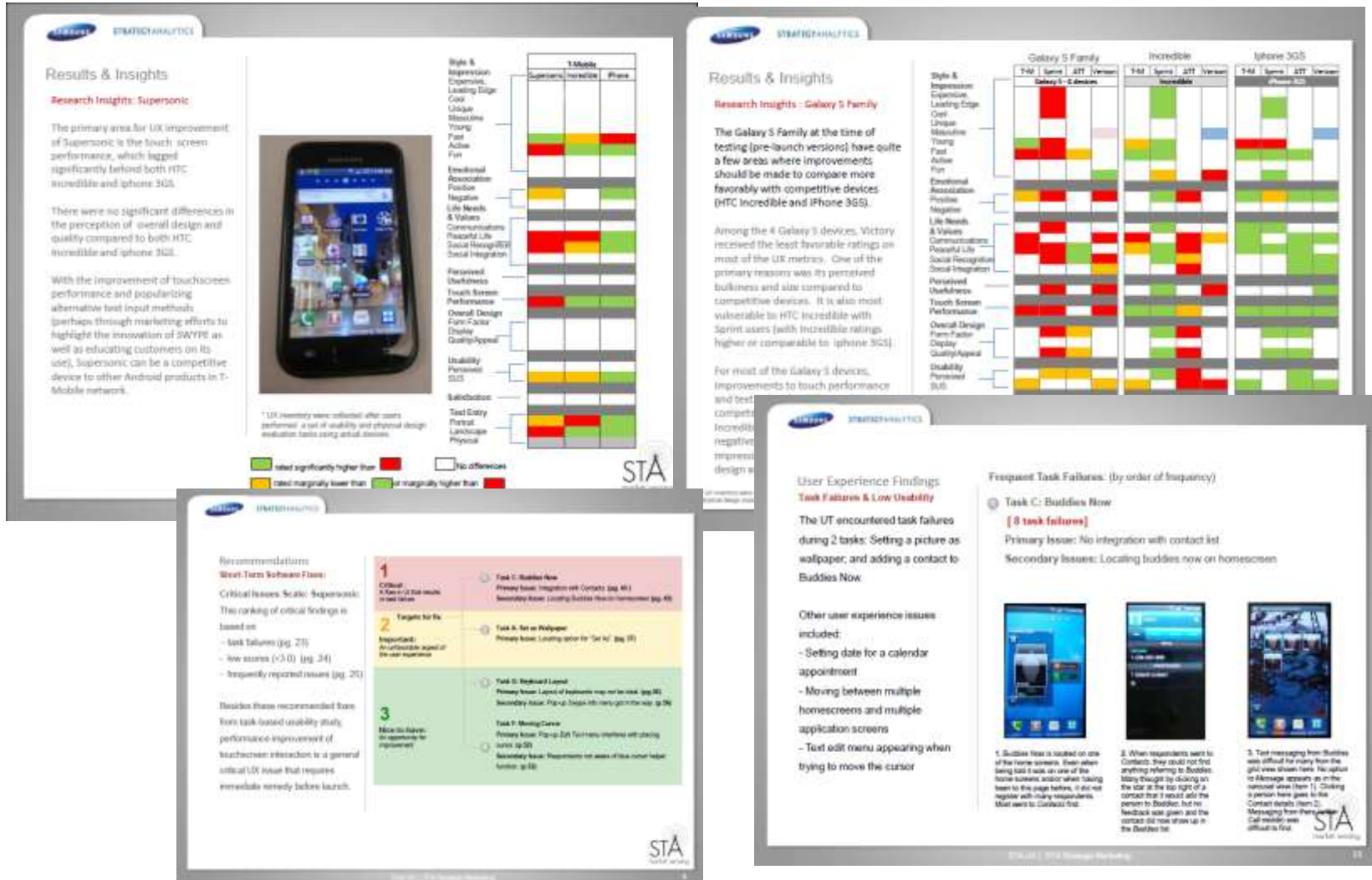


20. Galaxy S Device Product Experience: Samsung Atlas, HTC Incredible, iPhone 3GS



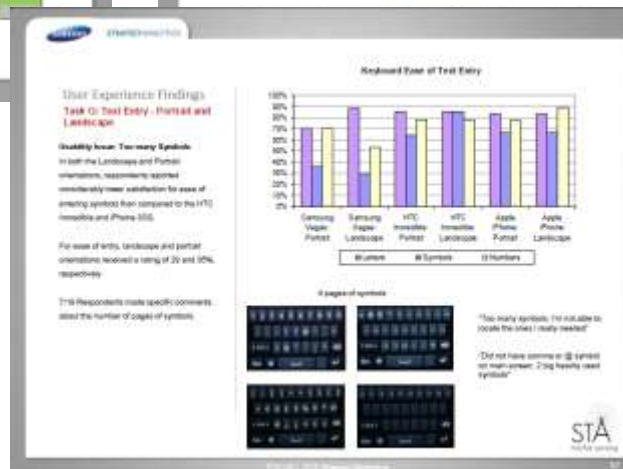
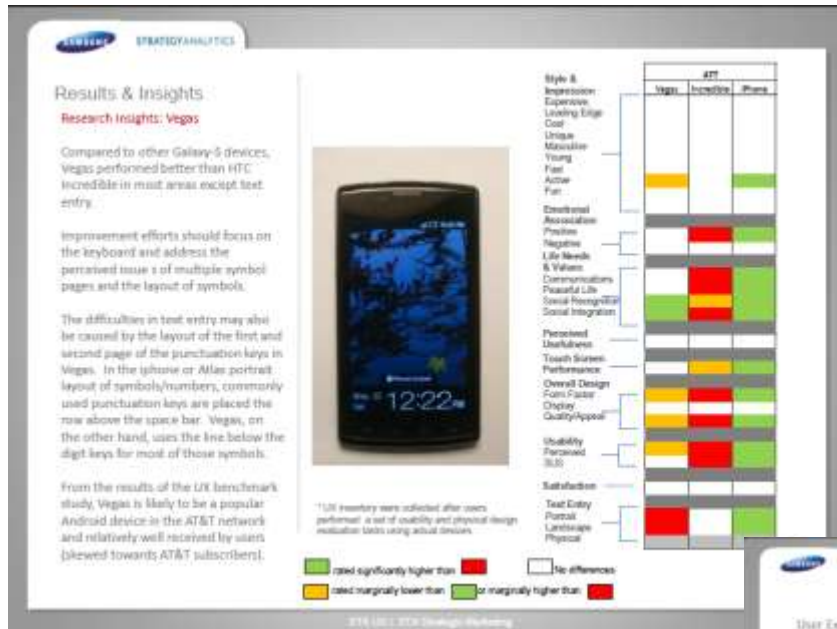
21. Galaxy S Device Product Experience: Samsung Supersonic, HTC Incredible, iPhone 3GS

Overview and Representative Results



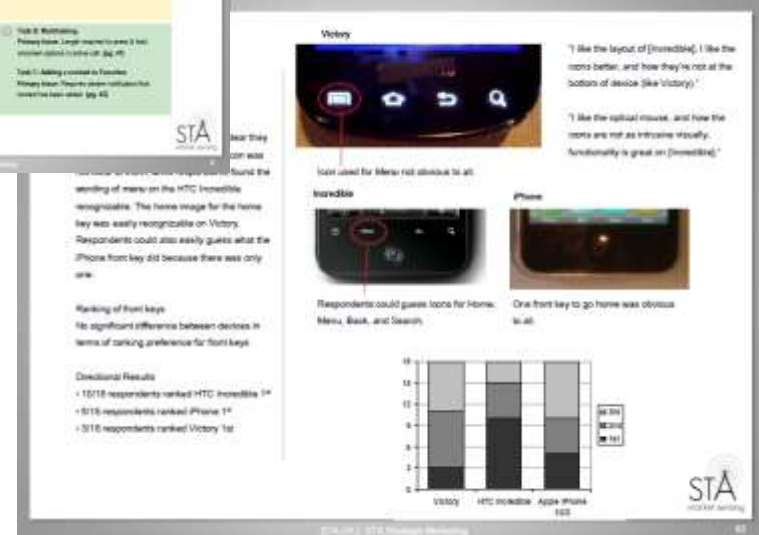
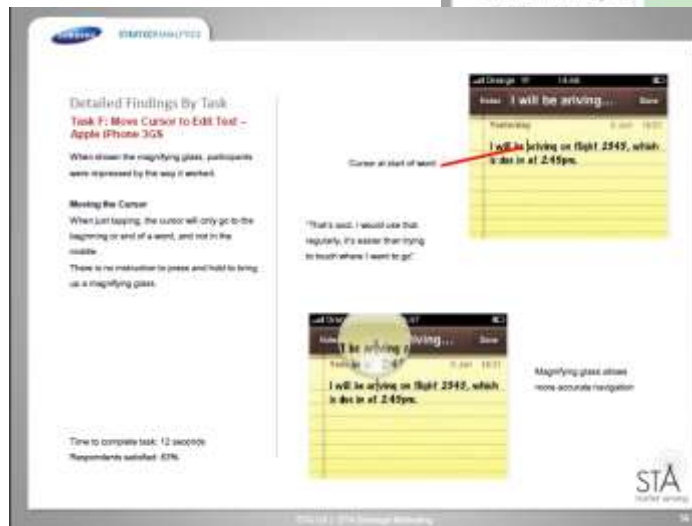
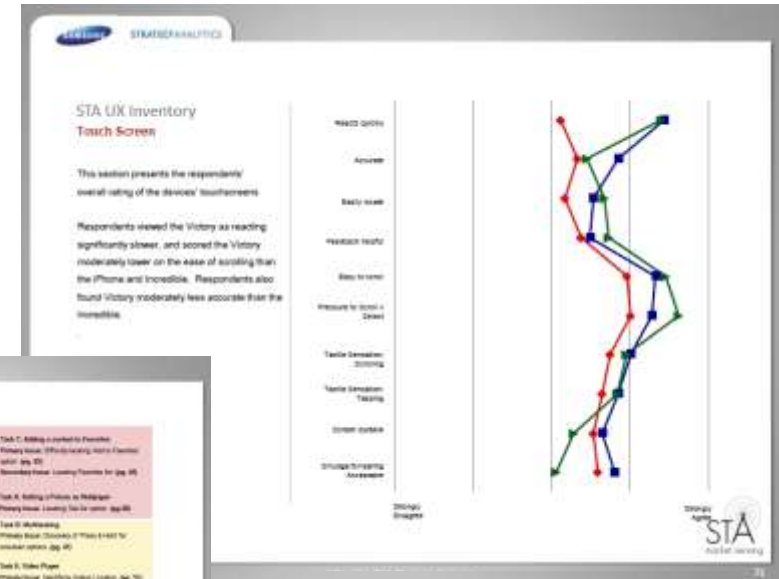
22. Galaxy S Device Product Experience: Samsung Vegas, HTC Incredible, iPhone 3GS

Overview and Representative Results



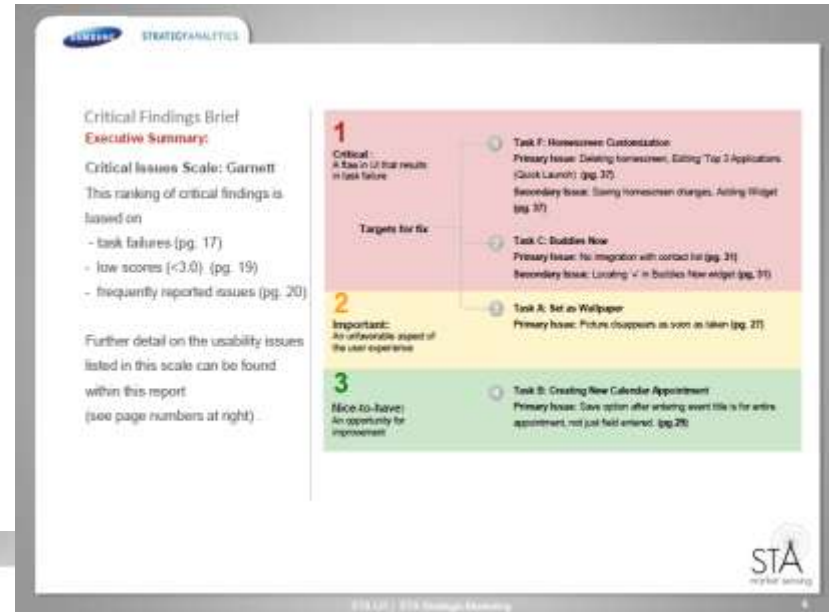
23. Galaxy S Device Product Experience: Samsung Victory, HTC Incredible, iPhone 3GS

Overview and Representative Results



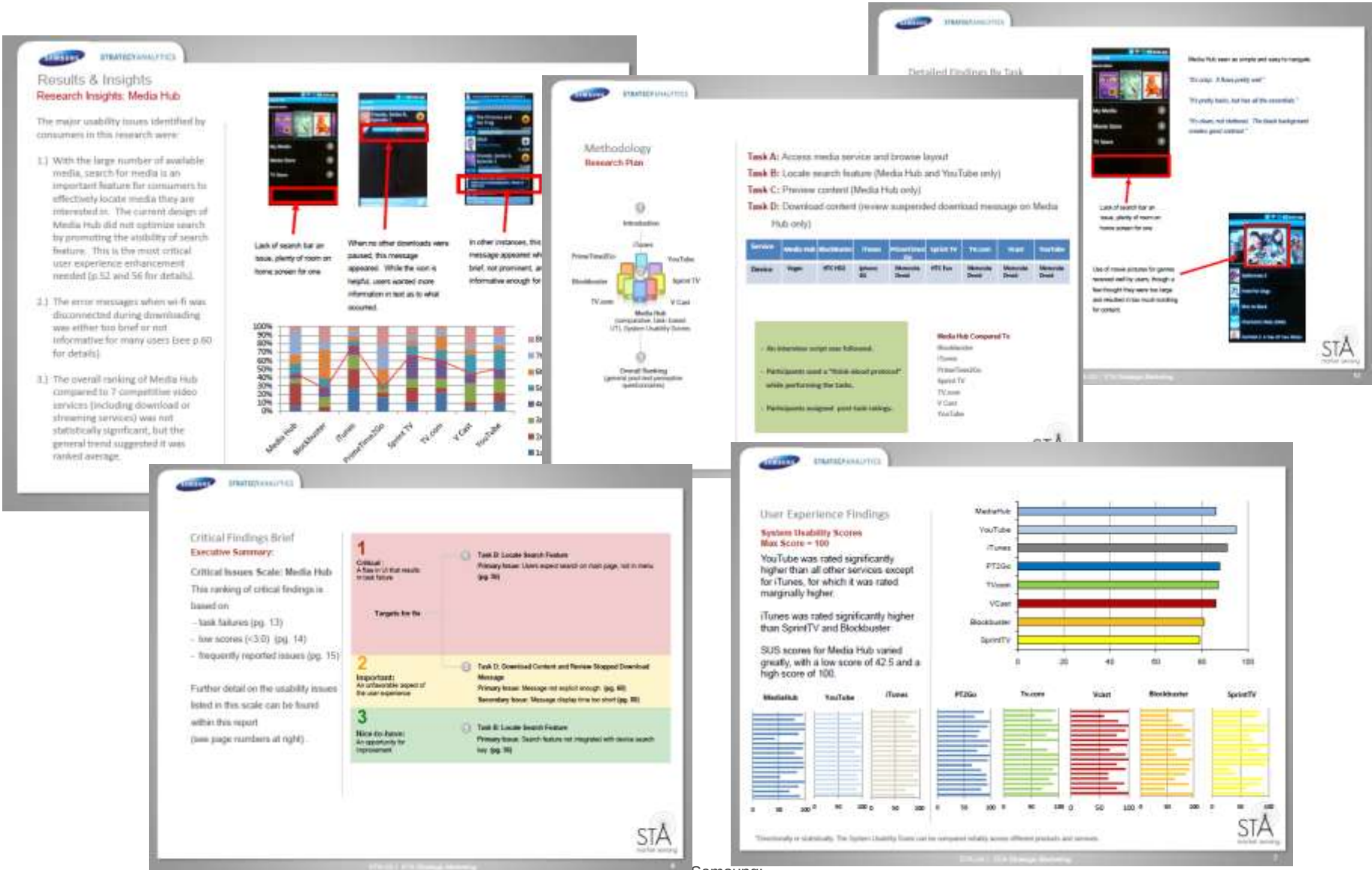
24. Garnett Usability Study (2ndround) & Competitive UX benchmark with HTC Evo

Overview and Representative Results



25. Samsung Media Hub – User Experience Evaluation

Overview and Representative Results



Overview and Representative Results



27. Sprint Rant3 Design Research

Overview and Representative Results

Executive Summary

Comparisons of Rant3 Overall Designs

- The overall design and comfort ratings were similar for both V and W samples.
 - Sample V received a 3.8 out of 5 for Design Attractiveness and 4.2 out of 5 for Comfortable to Hold in One Hand.
 - Sample W received a 3.3 out of 5 for Design Attractiveness and 4.2 out of 5 for Comfortable to Hold in One Hand.
- Preference for Sample V over Sample W was not significant, however, the trend is toward preference for Sample V.
 - Overall preference: 48 preferred V, 36 preferred W.
- There were no differences in any attractiveness ratings or preference ranking as a function of gender (M/F) or respondent age (three categories: 13-24, 25-35, 36 and older).

Sample V Sample W

Introduction

Methodology

- The goal of this study was to compare and understand what customers' preference between two different overall design visually and while using them. And also comparing three keyboard designs and usability.
- Data collection were conducted in Lasoville, TX (N=84, 46 males and 38 females).
- By working in small groups of 2 to 3, the respondents completed the following tasks:
 - Observed the design of the two pink Rant3 samples (V and W).
 - Rated overall design and feel in hand of V and W.
 - Noted preference between V and W.
 - Completed a typing task on all 3 samples (V, W, and S).
 - Rated each keyboard on a number of dimensions.
 - Rated ease of use and over satisfaction for V, W, and S.
 - Ranked preference among all 3 keyboards (V, W, and S).
 - Rated the level of interest of solar cell option.

Session Activities

Notating

Task timing

1 Moderator

2-3 Respondents in a group

Overall Design Comparison

Physical Keyboard Design Comparison

Post-Test Ranking

Keyboard Design Comparison

Ease of Typing and Satisfaction

- The overall rank of sample S-keyboard was significantly higher than sample V for all (and male) respondents.
- Satisfaction with sample S and sample W keyboards were significantly higher than sample V keyboard for all (and male) respondents.
- Ease of Typing with sample S and sample W keyboards was significantly higher than sample V keyboard for all (and male) respondents.

Rank distribution

Best Performance	Sample V			Sample W			Sample S		
	Total	M	F	Total	M	F	Total	M	F
1st	20	8	12	31	16	15	36	24	11
2nd	30	20	10	27	15	12	25	13	16
3rd	34	20	14	26	17	11	23	11	11

	Rating (5 is highest)		
Time of Typing (Min)	Sample V	Sample W	Sample S
All (N=86)	3.37	3.70	3.88
Male (N=46)	3.18	3.71	3.47
Female (N=40)	3.52	3.65	3.47

	Satisfaction		
Time of Typing (Min)	Sample V	Sample W	Sample S
All (N=86)	3.31	3.63	3.80
Male (N=46)	3.23	3.68	3.71
Female (N=40)	3.18	3.55	3.50

Green and red text indicate values that differ significantly from each other. (Green = significantly higher, Red = significantly lower)

Tactile feel of key separation

Keyboard Design Comparison

- Both Sample W and Sample S Keyboards scored significantly higher than Sample V keyboard for all (and female) respondents in tactile feel of separation of keys.
- The score for Sample S keyboard was marginally higher than Sample V keyboard for male respondents.
- It is interesting to note the gender differences between the tactile perception of the separation of keys.

Sample V Sample W Sample S

	Sample V	Sample W	Sample S
Time of Separation (sec)	3.87	4.20	4.44
Male (N=46)	3.80	4.31	4.50
Female (N=40)	3.94	4.02	4.38

Green and red text indicate values that differ significantly from each other. (Green = significantly higher, Red = significantly lower)

28. Tablet Device Evaluation: Apple iPad, Dell Streak, Samsung P1

Overview and Representative Results



30. Competitive UX Design & Paradigms - Samsung Captivate, Motorola Droid-X, Apple iPhone 4 & 3GS

Overview and Representative Results

Executive Summary

Fast Facts...

1. Samsung Captivate is highly competitive with iPhone 4. There were no statistical differences between overall UX index of Captivate and iPhone 4. Captivate was perceived positively by respondents in the clarity, color and size of the screen and its sleek, modern, aerodynamic and thin design.

2. Samsung Captivate scored significantly higher than Droid-X in SUS (System Usability Scores) and overall design UX index.

3. Samsung Captivate scored marginally higher than iPhone 4 in the ease of Symbols entry. The result was consistent with previous comparative study between iPhone 4 and Samsung Captivate.

5

4

3

2

1

0

Captivate Droid-X iPhone 4

STA overall UX Index

5

4

3

2

1

0

Captivate Droid-X iPhone 4

STA Usefulness/Life-Style Index

100

90

80

70

60

50

iPhone 4

Droid-X

Captivate

System Usability Scale*

5

4

3

2

1

0

Captivate Droid-X iPhone 4

STA Design Index*

Statistically significant at alpha = 0.05 (two-tailed)

Results & Insights

Competitive Design & UX Paradigm

A competitive Design and UX benchmark study was conducted with Samsung Captivate, Motorola Droid-X, iPhone 4 and iPhone 3GS. The purpose of this research was to measure any important design and UX factors that would effect users' overall experience and purchase decisions of their smartphones.

Each device was used 2 device sessions (90-120) consisting of 2 sections:
1.1 Respondents partially simulated their smartphone companion process when they dropped at a CS retail store. Each smartphone was evaluated by exploring its design, features and functions with minimal guidance from the researcher in-store carrier. The physical design and UX aspects initial experience were rated. The studies were then ranked according to overall design and performance.

2.1 A researcher-led UX walkthrough of each device (in random order) was then conducted, highlighting common features of 4 devices including iPhone 3GS (e.g., multi-touch, icons, home screen navigation, notifications, customization) as well as unique features (e.g., touch map behavior). Typing performance and satisfaction using the preferred presentation of the virtual QWERTY keyboard of each device was captured. Respondents completed the STA UX inventory after interacting with each device. After respondents completed all devices, they were asked to rank their preferences of the device again.

STAIR

Research

STAIR

Research

User Experience Findings

Perceived Experience: Motorola Droid-X

This section reported aspects of the phone that elicited feedback by respondents. These represented perceived experience by individual respondent from user's viewpoint.

Perceived Positive Experience

Perceived Negative Experience

Executive Summary

Improvements & Action Items...

1. The absence of Camera Flash of Captivate had the highest frequency of negative comments of the device. Since many usage scenarios of camera are indoor with subdued lighting, users deem flash is an important factor in their perception of the device and their purchase decision consideration. It is recommended that camera flash should be included as a standard feature in future smartphone design.

2. The perceived durability and quality of Captivate lagged behind iPhone 4. Further research of the factors leading to the different perception of quality and durability will be required (e.g., design, material, feature, weight etc.). This will be explored in relation to the research question of the factors contributing to the perception of a "premium" product and the effect of brand vs. other factors leading to such perception.

3. The size of the device was perceived to be too large for both 1. hand usage and pocketability. Future design should strive to keep the size of the display while further reducing the form factor of the device.

4. The form factor of Droid-X (with 4.3" display) was deemed too large by respondents. Future design for smartphones with 4.3" or larger screens should be evaluated carefully to ensure users' acceptance of display and physical size of the device.

SAMSUNG

Confidential & Proprietary to Samsung:
Do not disclose & distribute outside Samsung

STA Design + Technology Research | Vision & Goals 33

29. QWERTY / Touch Preference: Overview and Representative Results

Key Learnings

- There is a positive relationship between phone/screen size and perceived ease of using a Virtual Keyboard on a Pure Touch handset. Users perceive that virtual keyboard is easier to use on a device with a larger touch screen.
- Overall, 3.7" or 4" is the screen size at which using a Virtual QWERTY keyboard becomes comparable to a Physical QWERTY for both Physical QWERTY and Pure Touch users
 - however, 17% of Physical QWERTY users maintain that a Touch keyboard is not comparable to a Physical keyboard at any size
- Across all user groups (12-key, Physical QWERTY, and Pure Touch), respondents strongly prefer the Physical keyboard at smaller screen sizes (2.8" and 3.2"), but this preference either dissipates at the 3.7" screen size (12-key and Physical QWERTY users) or reverses such that Touch is preferred (Touch users)
- 3.7" and 4" handsets are the most preferred screen sizes among the mix of devices with different screen sizes (2.8", 3.2", 3.7", 4" and 4.3"); however, Physical vs. Touch preference depends on current phone ownership
 - 12-Key users are more likely to prefer the 4" Physical QWERTY handset over all other phones
 - Physical QWERTY users are more likely to prefer the 3.7" Touch over all other phones
 - Touch users are more likely to prefer 3.7" and 4" Touch over all other phones

Recommendation

- For Touch phones, 3.7" and/or 4" are the optimal screen sizes to appeal to both current Touch users and Physical QWERTY users
- However, we should not eliminate all Physical QWERTY phones because:
 - there is a small but loyal Physical QWERTY market (~17% of current QWERTY users)
 - Physical keyboards may appeal to 12-key users seeking to upgrade

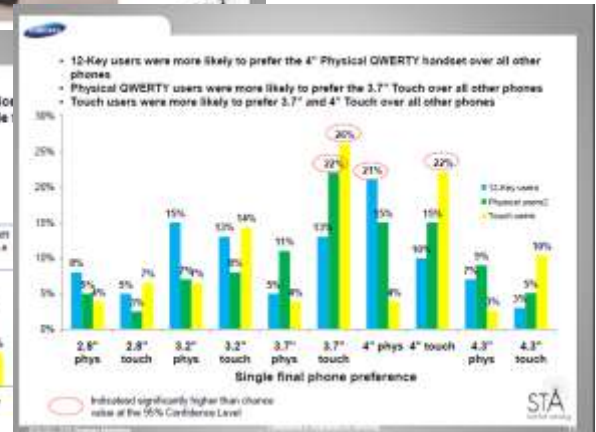
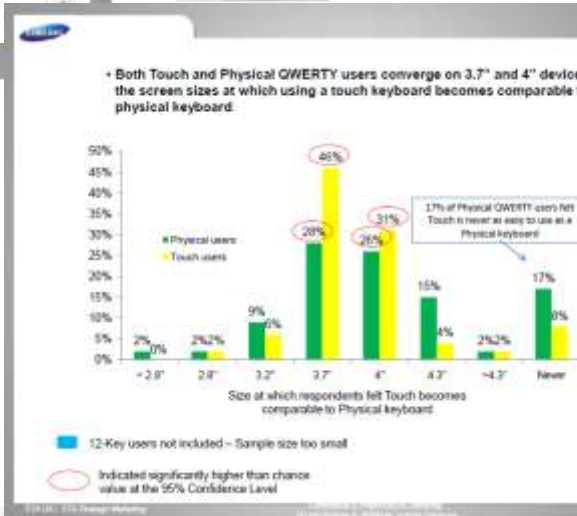
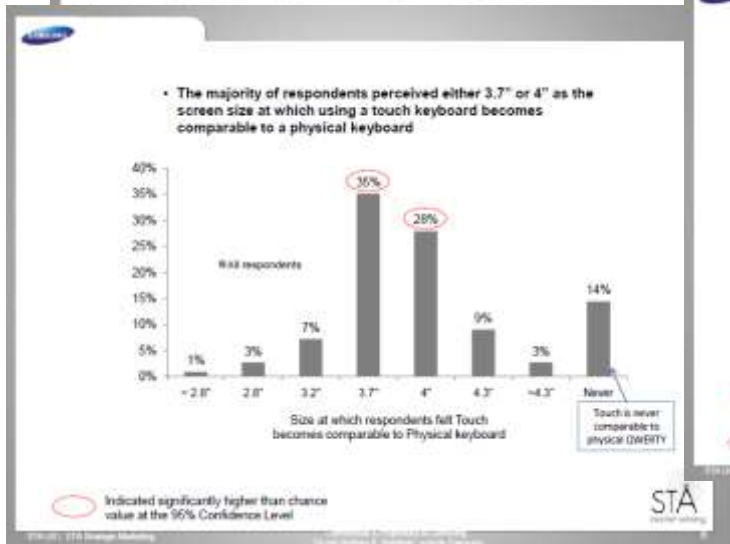
2.8"

3.2"

3.7"

4.0"

4.3"



30. Competitive UX Design & Paradigms - Samsung Captivate, Motorola Droid-X, Apple iPhone 4 & 3GS

Overview and Representative Results

Executive Summary

Fast Facts...

1. Samsung Captivate is highly competitive with iPhone 4. There were no statistical differences between overall UX index of Captivate and iPhone 4. Captivate was perceived positively by respondents in the clarity, color and size of the screen and its sleek, modern, aerodynamic and thin design.

2. Samsung Captivate scored significantly higher than Droid-X in SUS (System Usability Scores) and overall design UX index.

3. Samsung Captivate scored marginally higher than iPhone 4 in the ease of Symbols entry. The result was consistent with previous comparative study between iPhone 4 and Samsung Captivate.

5

4

3

2

1

0

Captivate Droid-X iPhone 4

STA overall UX Index

5

4

3

2

1

0

Captivate Droid-X iPhone 4

STA Usefulness/Life-Style Index

100

90

80

70

60

50

iPhone 4

Droid-X

Captivate

System Usability Scale*

5

4

3

2

1

0

Captivate Droid-X iPhone 4

STA Design Index*

Statistically significant at alpha = 0.05 (two-tailed)

User Experience Findings

Perceived Experience: Motorola Droid-X

This section reported aspects of the phone that elicited feedback by respondents. These represented perceived experience by individual respondent from user's viewpoint.

Perceived Positive Experience

Perceived Negative Experience

Results & Insights

Competitive Design & UX Paradigm

A competitive Design and UX benchmark study was conducted with Samsung Captivate, Motorola Droid-X, iPhone 4 and iPhone 3GS. The purpose of this research was to measure any important design and UX factors that would effect users' overall experience and purchase decisions of their smartphones.

Each device was used 2 device sessions (94-100) consisted of 2 sections:
1.1 Respondents partially simulated their smartphone companion process when they dropped at a CS retail store. Each smartphone was evaluated by exploring its design, features and functions with internal guidelines from the customer in feature card. The physical design and UX aspects initial experience were rated. The studies were then ranked according to overall design and performance.

2.1 A researcher-led UX walkthrough of each device (in random order) was then conducted, highlighting common features of 4 devices including iPhone 3GS (e.g., multi-touch, icons, home screen navigation, notifications, customization) as well as unique features (e.g., touch map behavior). Typing performance and satisfaction using the preferred presentation of the virtual QWERTY keyboard of each device was captured. Respondents completed the STA UX inventory after interacting with each device. After respondents completed all devices, they were asked to rank their preferences of the device again.

STA

market strategy

Executive Summary

Improvements & Action Items...

1. The absence of Camera Flash of Captivate had the highest frequency of negative comments of the device. Since many usage scenarios of camera are indoor with subdued lighting, users deem flash is an important factor in their perception of the device and their purchase decision consideration. It is recommended that camera flash should be included as a standard feature in future smartphone design.

2. The perceived durability and quality of Captivate lagged behind iPhone 4. Further research of the factors leading to the different perception of quality and durability will be required (e.g., design, material, feature, weight etc.). This will be explored in relation to the research question of the factors contributing to the perception of a "premium" product and the effect of brand vs. other factors leading to such perception.

3. The size of the device was perceived to be too large for both 1. hand usage and pocketability. Future design should strive to keep the size of the display while further reducing the form factor of the device.

4. The form factor of Droid-X (with 4.3" display) was deemed too large by respondents. Future design for smartphones with 4.3" or larger screens should be evaluated carefully to ensure users' acceptance of display and physical size of the device.

STA

market strategy

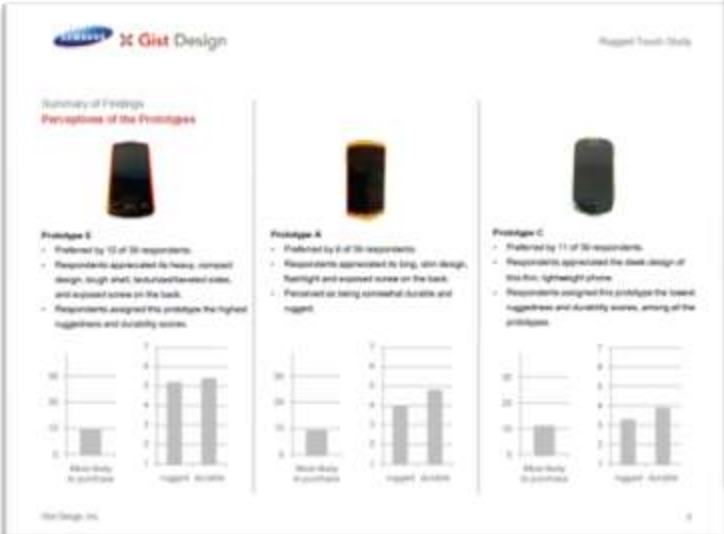
SAMSUNG

Confidential & Proprietary to Samsung:
Do not disclose & distribute outside Samsung

STA Design + Technology Research | Vision & Goals 35

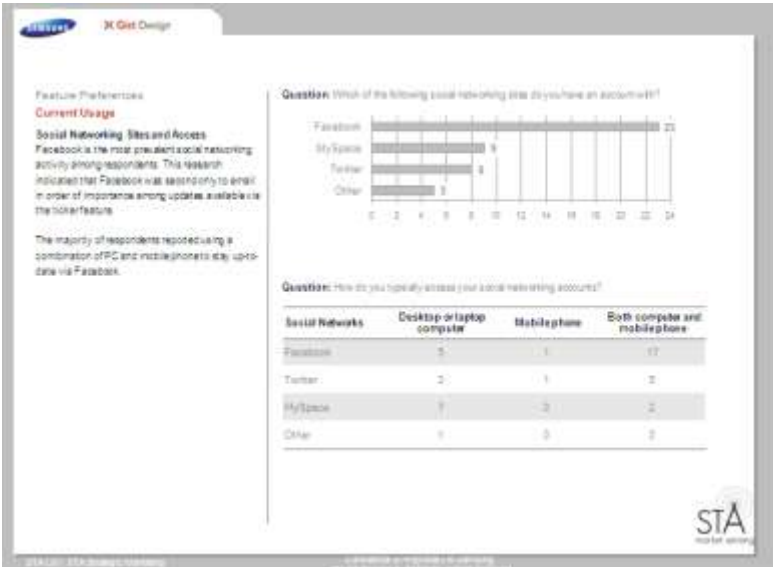
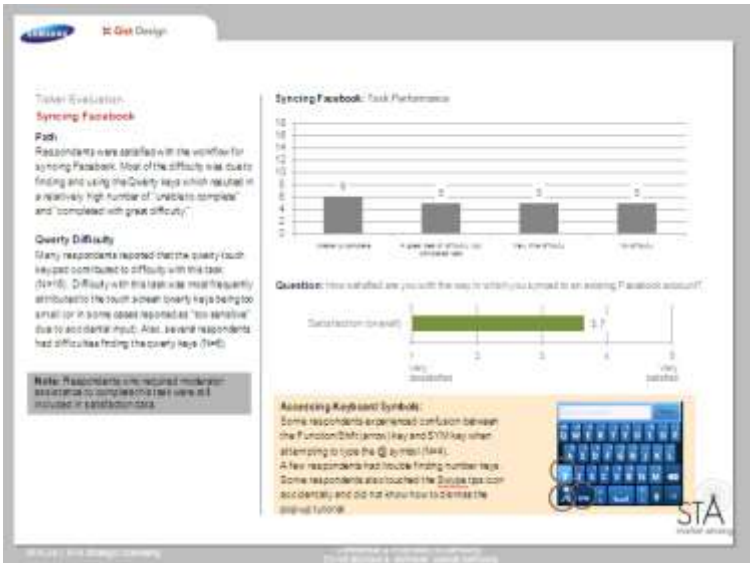
31. Rugged Touch Study

Overview and Representative Results



32. Continuum UX: Initial Use, Ticker & Grip Sensor

Overview and Representative Results



33. Competitive Physical Design: Samsung Stealth, HTC Evoand Motorola Droid-X

Overview and Representative Results

Executive Summary

Improvements & Action Items...

- Although the design of physical buttons (6 Android keys) on Stealth was well received, there is room for further improvement:
 - The physical buttons felt as if they could get stuck easily.
 - The material of the buttons was perceived to be cheap and flimsy by a few respondents.
 - Physical buttons were too low for one-hand operations.
- A number of respondents thought the back of the Stealth could be scratched easily.
- Users generally did not like protrusion or hump/bump in the shape and form of the device. Such design should be avoided if possible.
- The size of the device was perceived to be too large for both 1-hand usage and pocketability. Future design should strive to keep the size of the display while further reducing the form factor of the device.
- The form factor of all the devices (with 4.3" display) was deemed too large by respondents. Future design for smartphones with 4.3" or larger screens should be evaluated carefully to ensure users' acceptance of display and physical size of the device. Periodic research on whether consumers will continue to accept larger and larger display despite physical ergonomic issues may shed light on the upper size limit of devices when they are used as mobile smartphones.

STA
market strategy

Results & Insights

Physical Design – Form Factor

User impressions across the devices showed that the overall size of all 3 devices were rated as significantly larger than "Just Right" by respondents.

The screen size of all the devices were also rated as significantly larger than "Just Right".

Only the weight of Samsung Stealth prototype was pegged as "Just Right".

	Motorola Droid X	HTC Evo	Samsung Stealth
Overall size	Too Large	Too Large	Too Large
Screen size	Too Large	Too Large	Too Large
Weight	Too Large	Too Large	Just Right

■ rated significant comparing to "Just Right" (p < 0.05)
■ rated "Just Right" (no significant differences from "Just Right")

User Experience Findings

Samsung Stealth

Buttons too low

Large, visible physical buttons

Large Screen

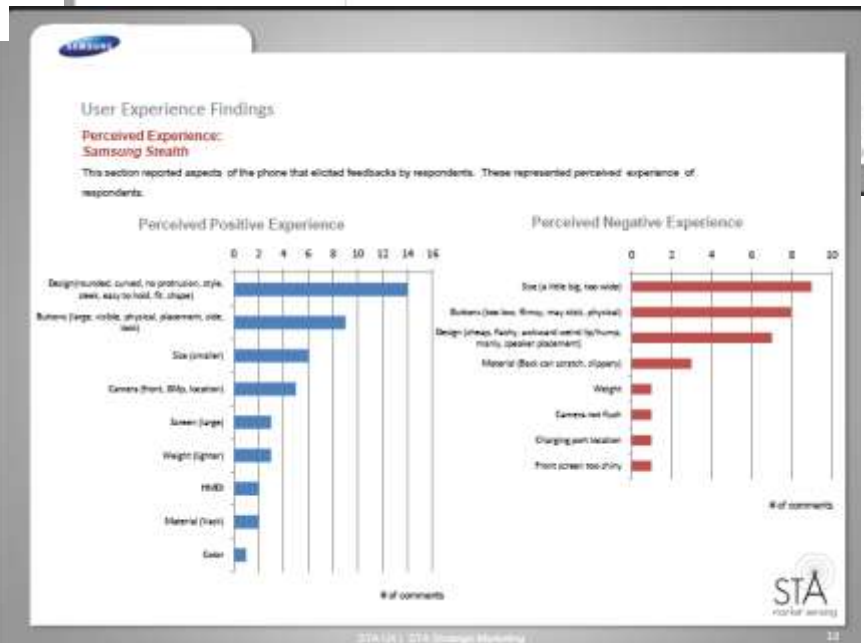
Rounded Edges

Buttons feel flimsy and may get stuck

Speaker placement too high

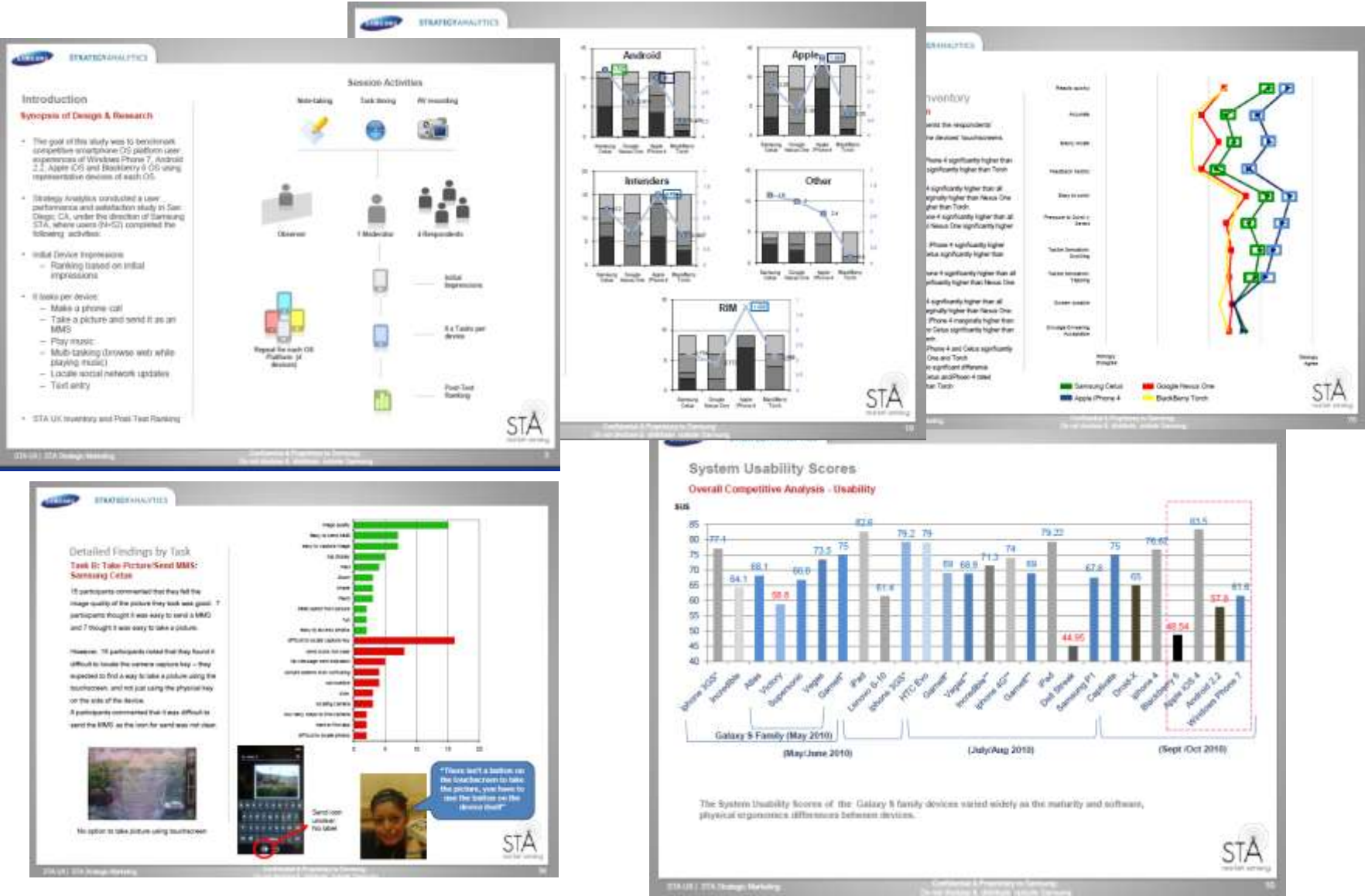
Back may get scratched easily

STA
market strategy



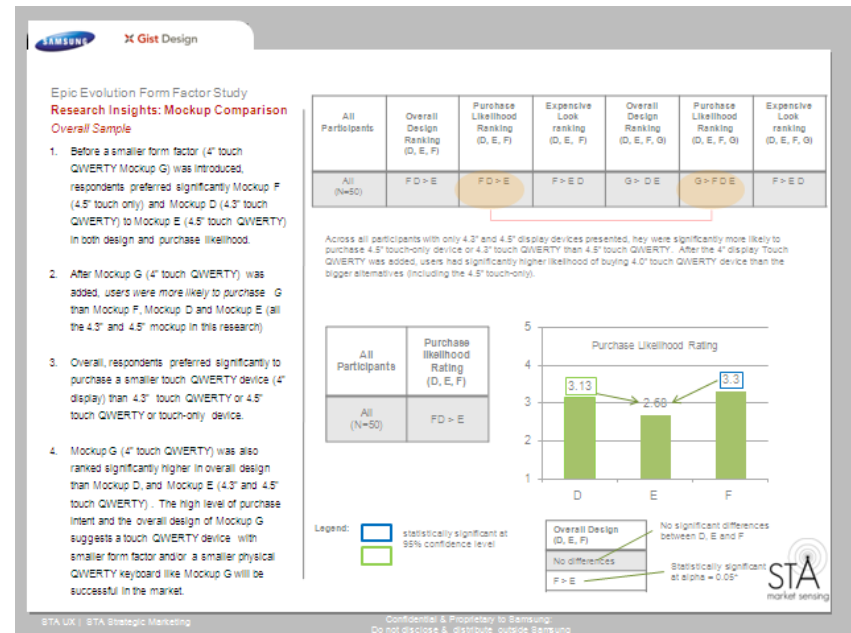
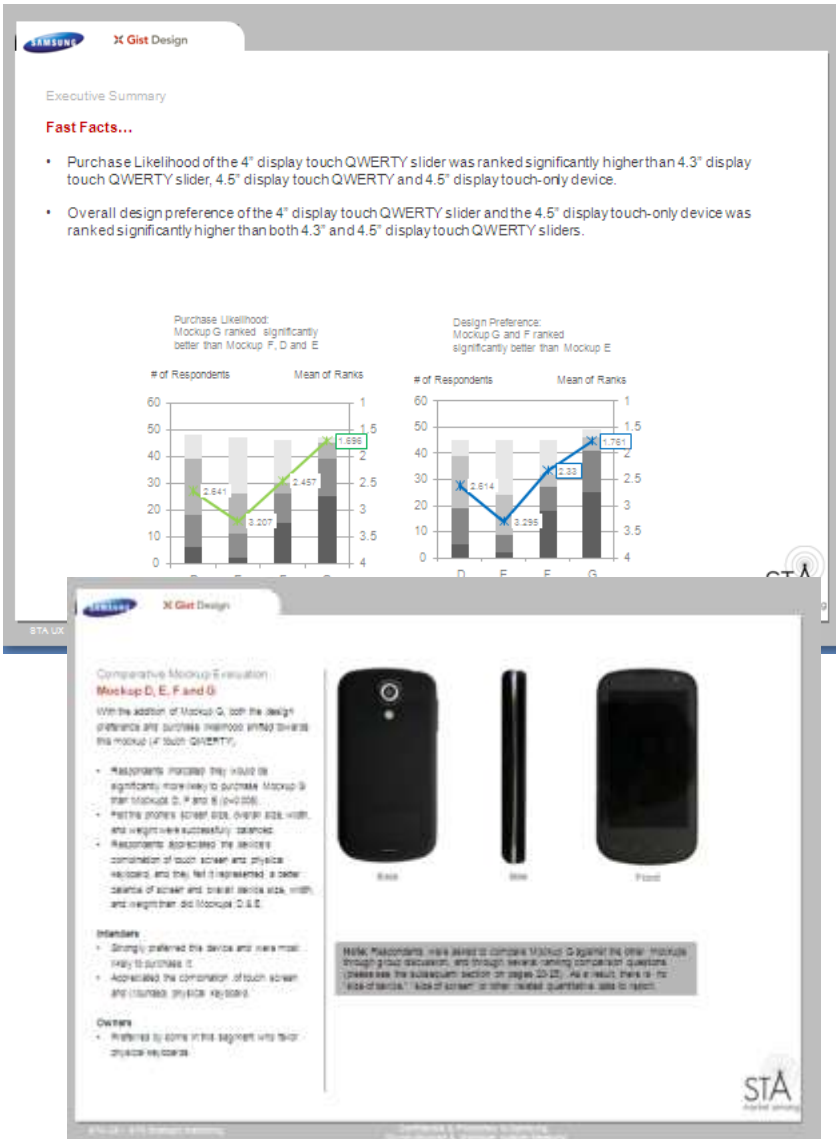
34. Competitive SmartPhone OS Platform UX – Windows Phone 7, Android 2.2, iOS 4 and Blackberry 6

Overview and Representative Results



35. Sprint Epic Evolution Form Factor Study

Overview and Representative Results



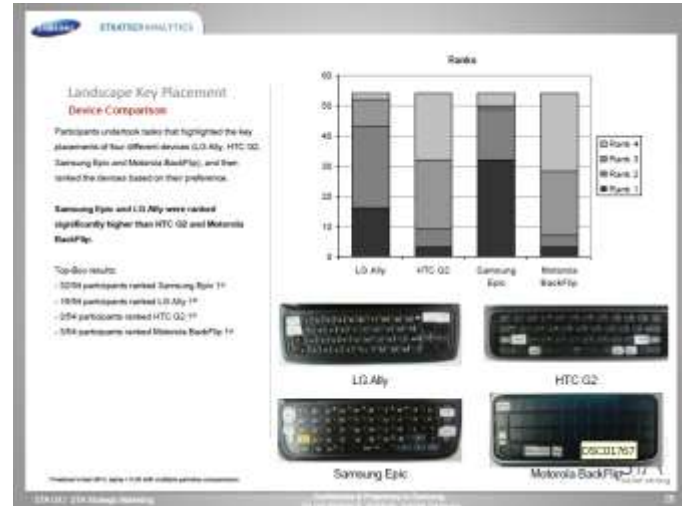
36. Android Key Placement Evaluation (Portrait & Landscape)

Overview and Representative Results

Executive Summary

Fast Facts...

- Users (N=54) prefer to have Android keys in the following configuration:
 - Home-Menu-Back-Search
- This result is supported by the following findings:
 - Significant preference of Home, Menu and Search keys in pre-evaluation placement of those keys in the respective locations
 - Significant satisfaction of placement of Menu key in second location to first location from the left (as exemplified by Evo vs. Epic).
 - Marginally significant satisfaction of overall Android key layout of Home-Menu-Back-Search vs. Menu-Home-Back-Search (as exemplified by Evo vs. Epic)
 - Significant preference of Home, Menu, Back and Search keys in post-evaluation placement of those keys in the respective locations
- The result is corroborated by previous research done in Jan/Feb 2010.
- The finding is similar to the Ideal Key Type Suggestions as reported in Android PUI Usability Testing done by HQ Mobile Design Group.
- Directional results suggest that the Android hard key layout for landscape QWERTY of Samsung Epic with the Android keys on the left and right side of the keyboard is preferred. Placement of Search key as the right lower key is significant.



Introduction

Synopsis of Design & Research

- The goal of this study (N=54, about 75% smartphone owners and 25% smartphone intenders) was to determine the best placement for Android keys both in portrait mode and in landscape mode with a slide out QWERTY keyboard. The keys evaluated were:
 - Back
 - Home
 - Menu
 - Search
- Participants placed keys where they initially thought were the ideal locations.
- Participants then went through a series of tasks that showcased different key placements on the following devices:
 - Portrait - HTC Evo, Samsung Epic, Motorola Droid (unmarketed)
 - Landscape - LG Ally, HTC G2, Samsung Epic and Motorola BackFlip (unmarketed)
- Participants provided satisfaction ratings for each configuration and ranked them in order of preference.
- After experiencing different devices, participants again placed keys in where they felt were the ideal locations.

Session Activities

Flowchart showing the session activities: Role-playing, AI recording, Observation, 1 Moderator, 2 Respondents, Initial Key Placement, 3 tasks (portrait, 5 tasks landscape), Post-Test Ranking, and Final Key Placement.

Executive Summary

Recommendations...

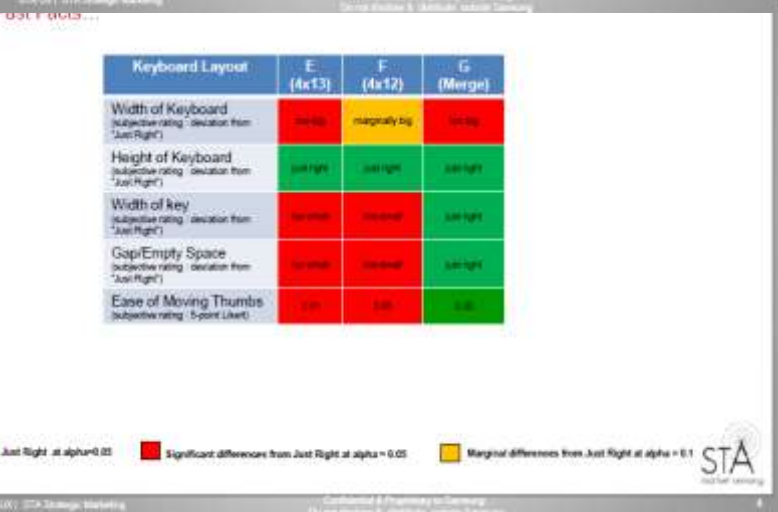
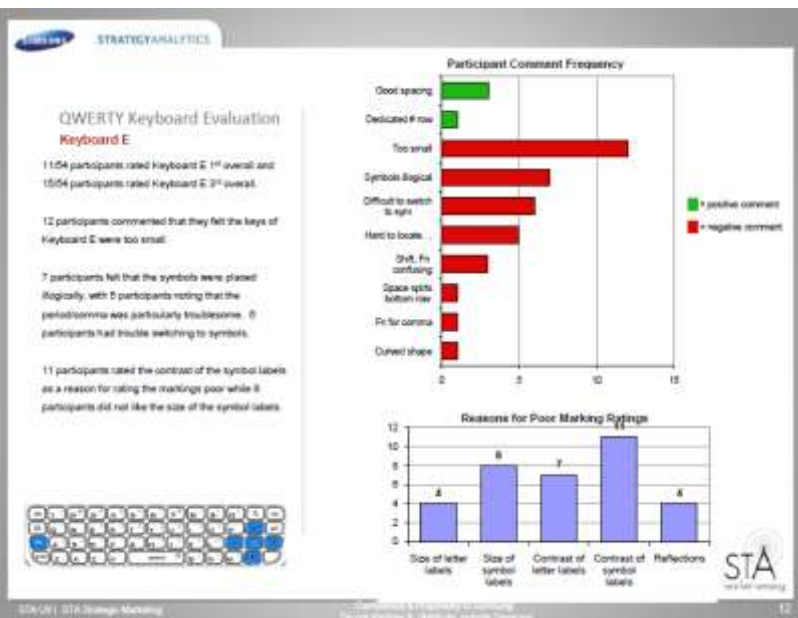
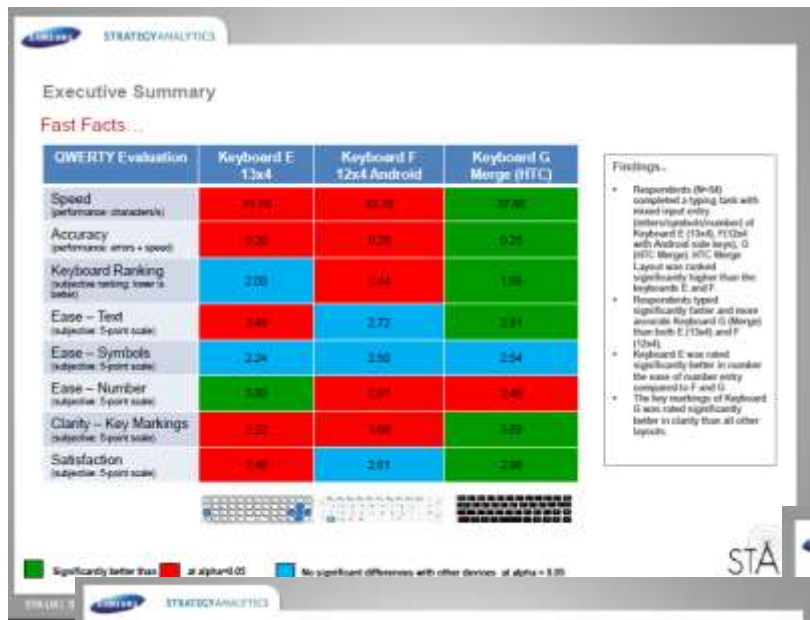
- From the behavioral research results conducted on smartphone owners and intenders, the following order of Portrait Android buttons is recommended:
 - Home-Menu-Back-Search.
- The Android hard key layout for landscape QWERTY of Samsung Epic with the Android keys on the left and right side of the keyboard is a recommended approach. Placement of Search key as the right lower key is also recommended.

Images showing the recommended key layouts for Portrait and Landscape modes.

37. QWERTY Keyboard Layout Evaluation Keyboard Layout

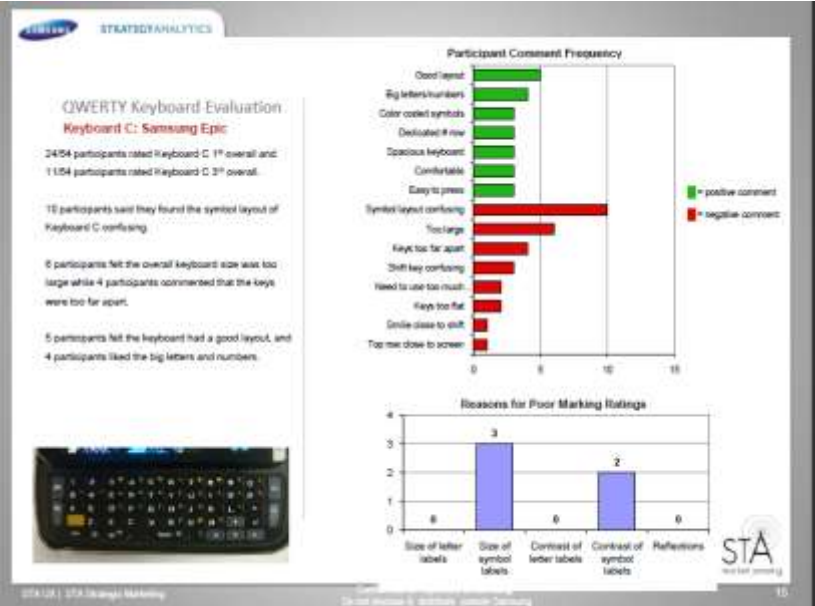
(Samsung 13x4 Layout, Samsung 12x4 Layout - side Android Keys, HTC Merge Layout)

Overview and Representative Results

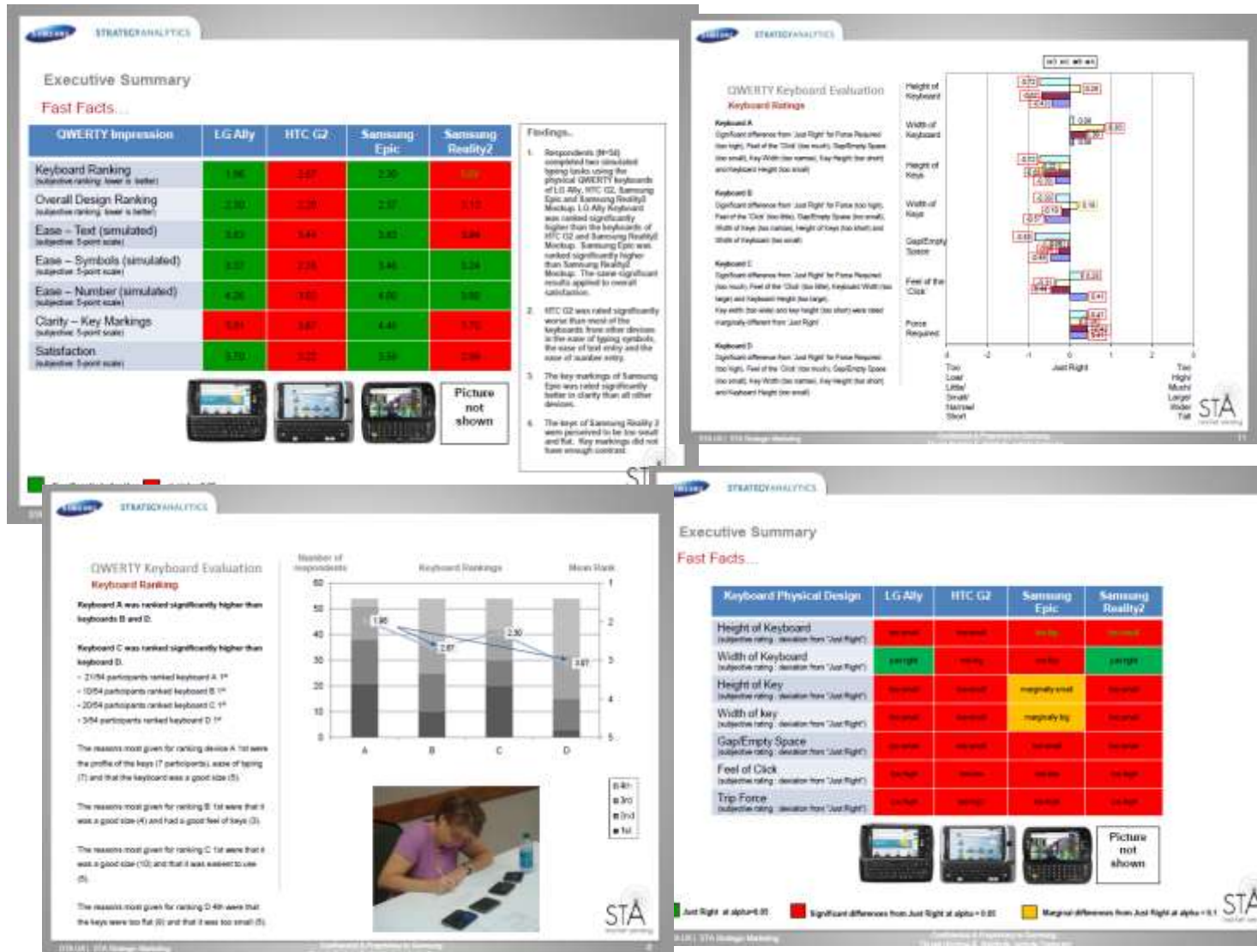


38. QWERTY Keyboard Layout Evaluation Keyboard Performance Benchmarking (LG Ally, HTC G2, and Samsung Epic)

Overview and Representative Results



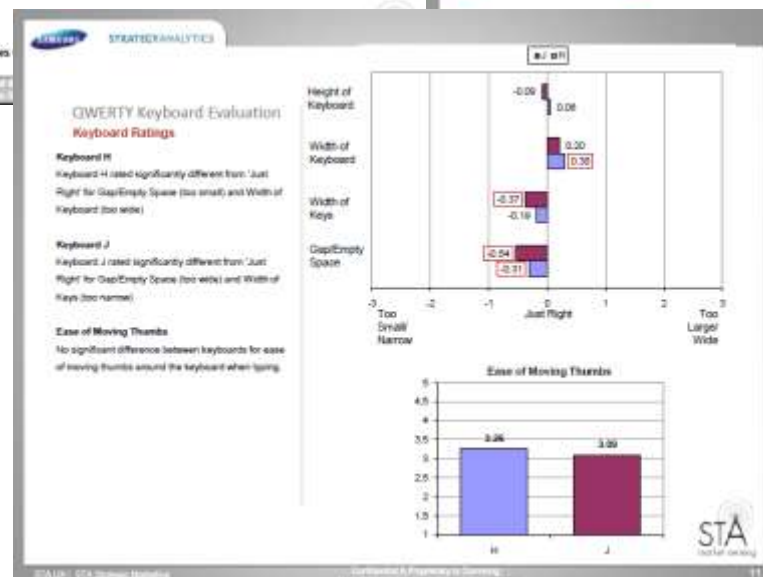
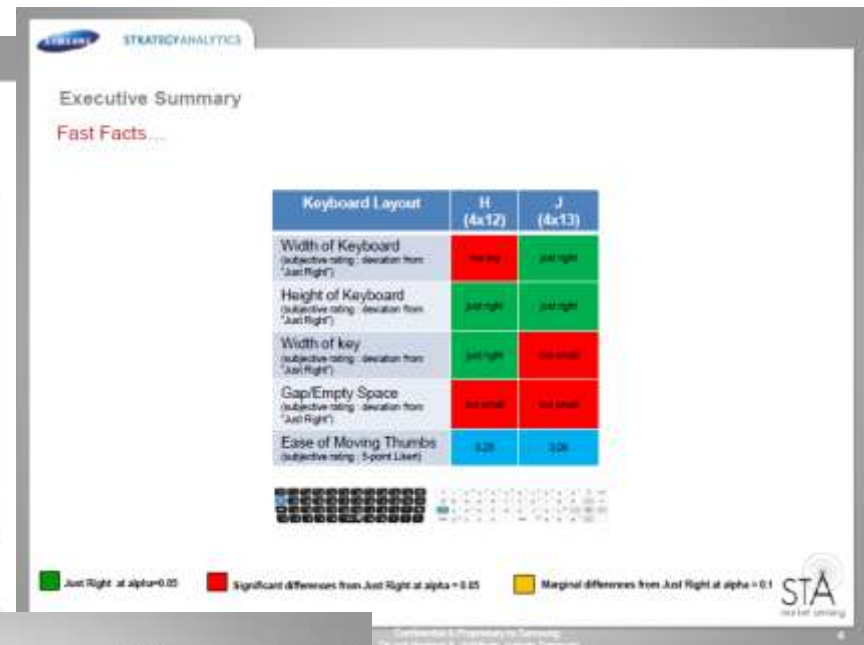
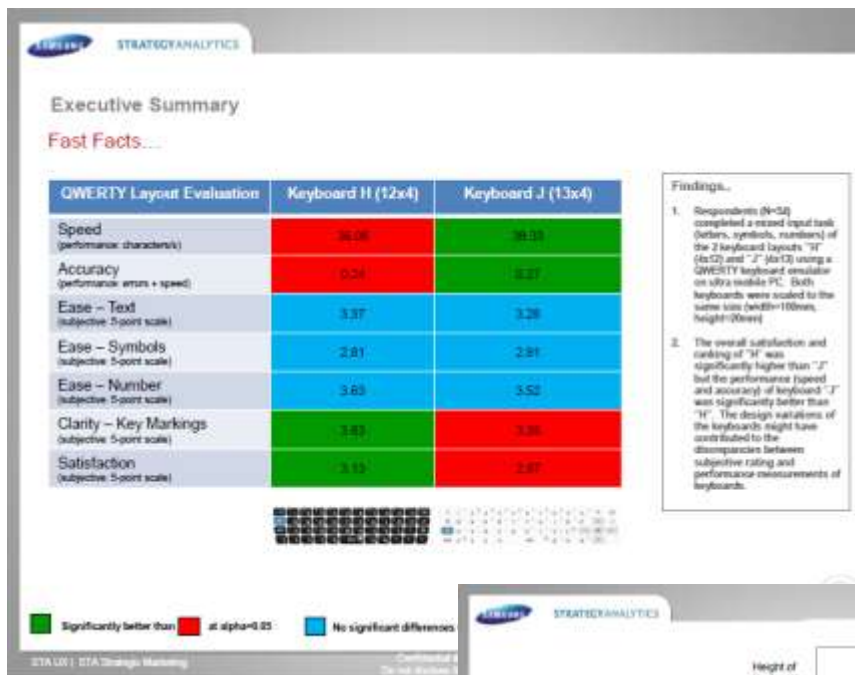
39. QWERTY Keyboard Layout Evaluation Keyboard Design Benchmarking (LG Ally, HTC G2, Samsung Epic and Samsung Reality2 Mockup) Overview and Representative Results



40 QWERTY Keyboard Layout Evaluation

Samsung Intensity Keyboards Alternative Layouts (13x4 and 12x4)

Overview and Representative Results



41 QWERTY Keyboard Layout Evaluation

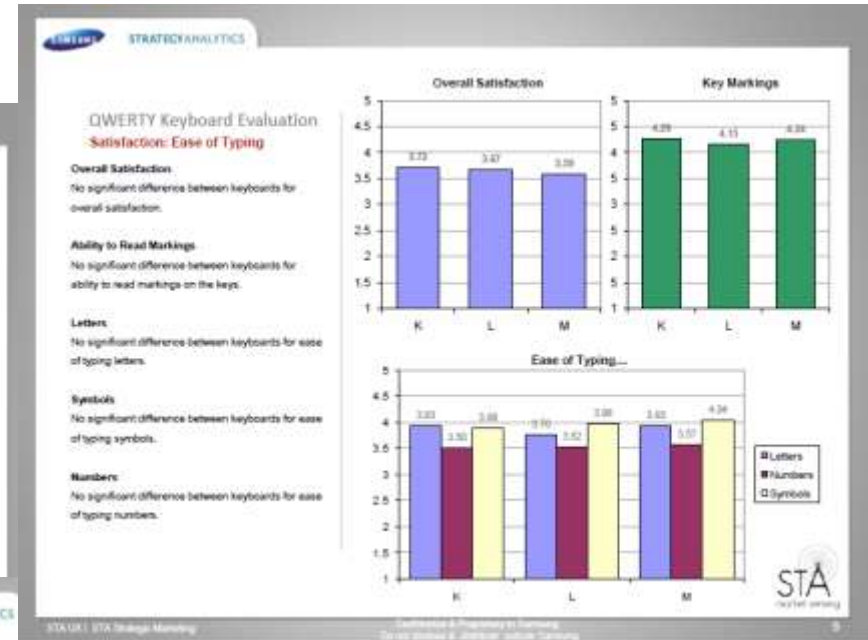
Samsung 5-line Keyboards in 4.5", 4.3" and 4" display (based on Epic)

Overview and Representative Results



Findings...

- Respondents (N=54) completed a typing task with mixed input entry (letters/symbols/numbers) of Keyboard K (4" display form factor), L (4.3" display form factor) and M (4.5" display form factor).
- No significant results except in the ease of moving thumbs or fingers during typing. Keyboard K was rated significantly better than Keyboards L and M.



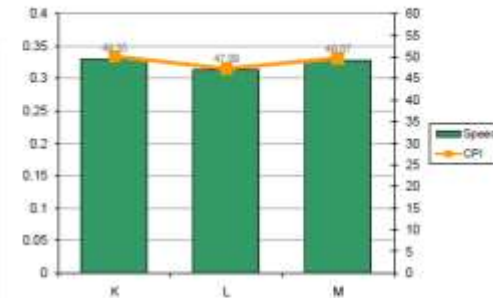
QWERTY Keyboard Evaluation
Keyboard Performance

Speed
No significant difference between keyboards for speed. Keyboard K rated marginally higher than Keyboard L for speed.

- Participants entered an average of 49.35 characters per minute using keyboard K.
- Participants entered an average of 47.06 characters per minute using keyboard L.
- Participants entered an average of 49.07 characters per minute using keyboard M.

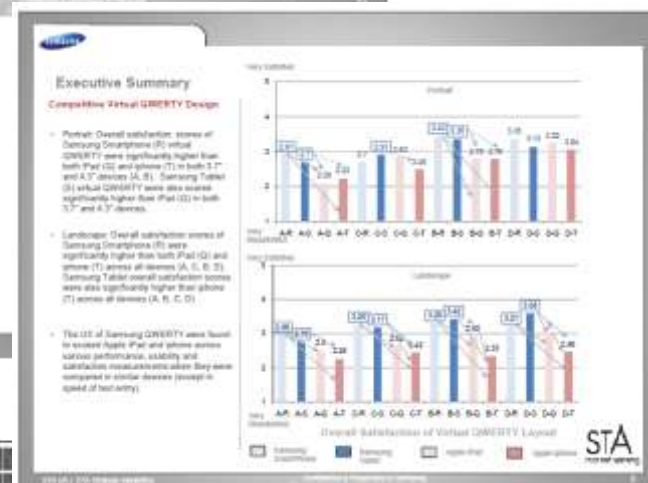
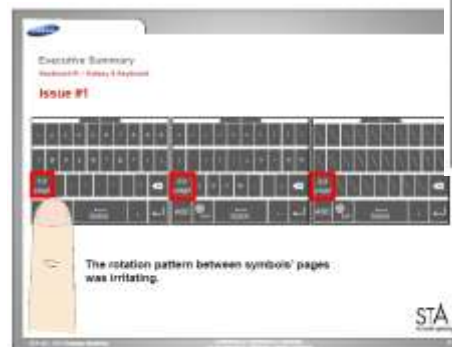
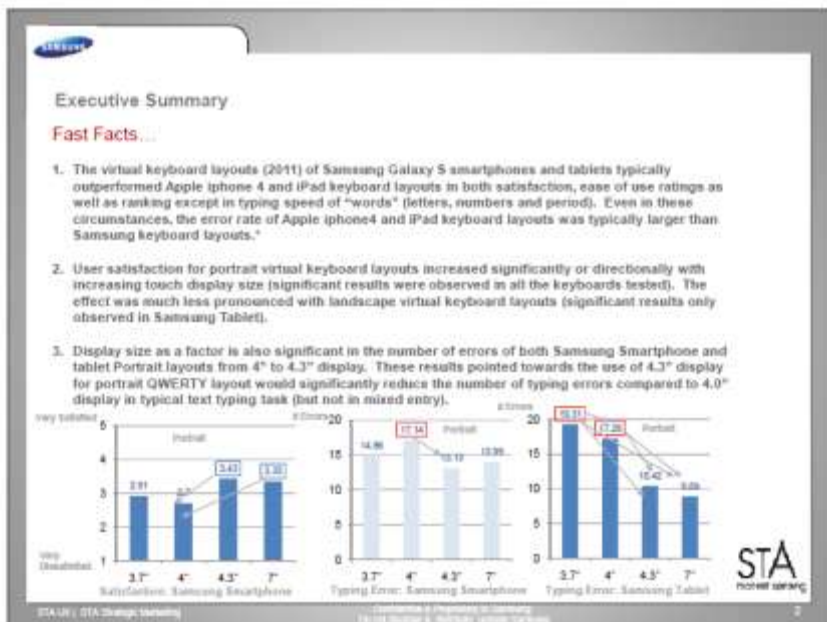
CPI
Keyboard K rated significantly higher than Keyboard L for CPI. Keyboard M rated marginally higher than Keyboard L for CPI.

- Participants scored an average CPI of 0.334 for keyboard K.
- Participants scored an average CPI of 0.315 for keyboard L.
- Participants scored an average CPI of 0.332 for keyboard M.

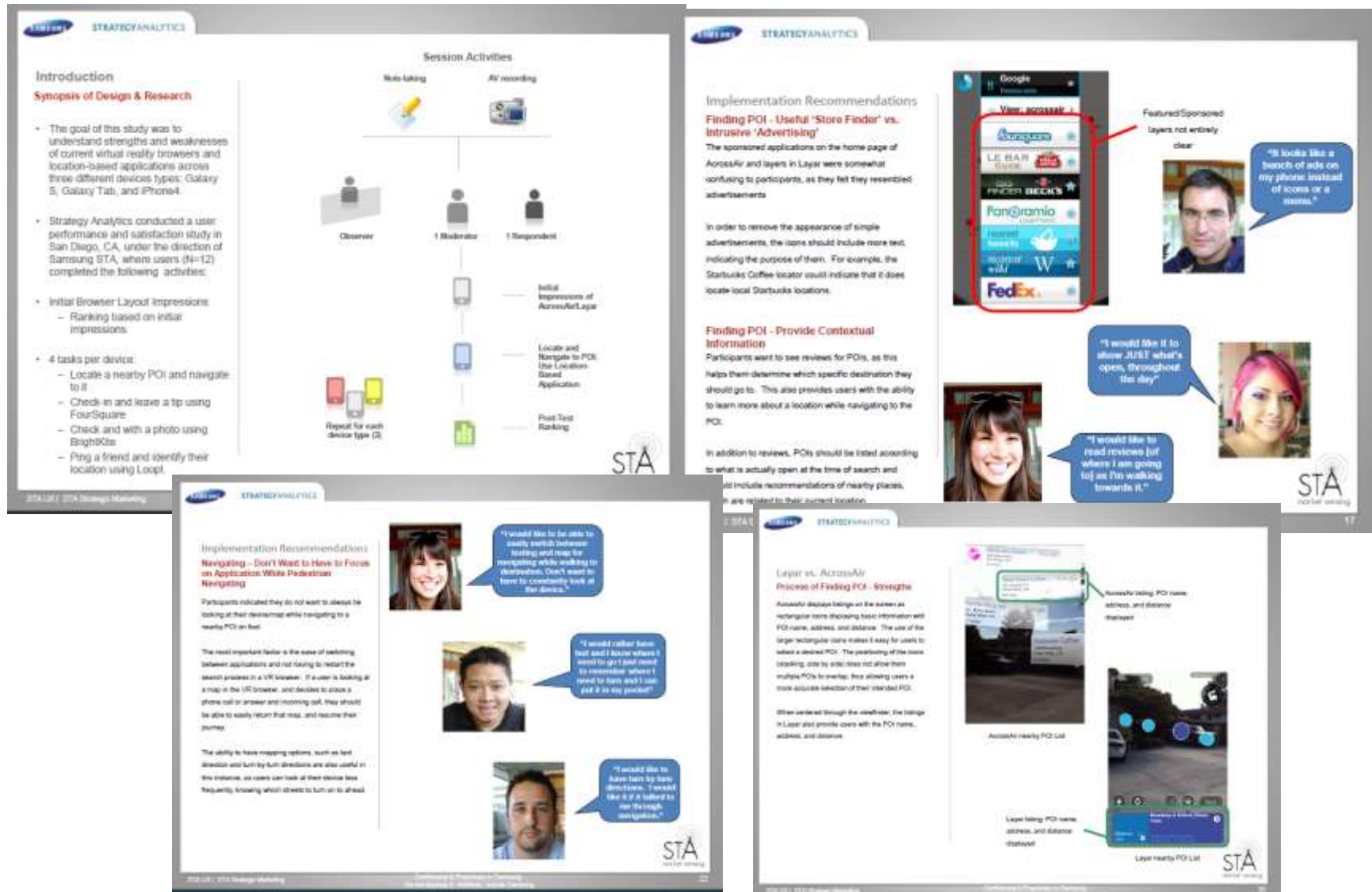


42. Mobile QWERTY Virtual Keyboard Design (performance & satisfaction)

Overview and Representative Results



43. Virtual Reality Browsers & Location-Based Applications: (Layar, AcrossAir, Loopt, BrightKite, FourSquare) Overview and Representative Results



Overview and Representative Results

